

A
PROJECT REPORT
On
**Modelling And Simulation of UPQC For
Power Quality Improvement**

Submitted in partial fulfilment for the award of the degree
Of
BACHELOR OF TECHNOLOGY
In
ELECTRICAL ENGINEERING

Submitted by

SANISHISHA KHYRIEMMUJAT	(D/23/EE/202)
SWARUP KAR CHAUDHARI	(D/22/EE/105)
GRACEFULLY SYNGKLI	(D/22/EE/103)
DAPMAIN PAPANG	(D/22/EE/018)



Under The Guidance of

Dr. M. Deben Singh
Associate Professor

North Eastern Regional Institute of Science and Technology
(Deemed to be University Under the Ministry, Govt. Of India)
Nirjuli (Itanagar), Arunachal Pradesh – 791109
Session (Jan-May 2026)



पूर्वोत्तर क्षेत्रीय वर्वज्ञान एर्वम प्रौद्योविकी संस्थान

NORTH EASTERN REGIONAL INSTITUTE OF SCIENCE AND TECHNOLOGY

(Deemed to be University u/s 3 of the UGC Act 1956)

Under the Ministry of Education, Govt. of India

वनरजुली - ७९११०९ (ईटानिर)
अरुणाचल प्रदेश, भारत

Nirjuli - 791109 (Itanagar)
Arunachal Pradesh, India

Certificate of Approval

This is to certify that the project report entitled “*Modelling And Simulation of UPQC For Power Quality Improvement*” being submitted by Ms. Sanishisha Khyriemmuja (Roll No:D/23/EE/202), Mr. Swarup Kar Chaudhari(Roll No:D/22/EE/105), Ms. Gracefully Syngkli (Roll No: D/22/EE/103), and Mr. Dapmain Papang (Roll No:D/22/EE/018) in partial fulfilment for the award of the Degree of Bachelor of Technology in Electrical Engineering of the North Eastern Regional institute of Science and Technology (NERIST) is based on the theoretical and simulation studies carried out by them during the academic semesters, July-December, 2025 to January-May, 2026 in a manner satisfactory to warrant its acceptance as a pre requisite for the award of the Degree in Electrical Engineering of the Institute.

It is understood by this approval that the undersigned does not endorse or approve any statement made, opinions expressed, or conclusion drawn therein, but only for the purpose for which it has been submitted.

(Dr. M. Deben Singh)
Associate Professor
Project Supervisor

(Prof. T. Vimal Prakash Singh)
Associate Professor
Project Coordinator

(Prof. Radak Blange)
Head of Department
Associate Professor

Phone : (0360) 2257401-11(O)
Fax: (0360) 2257872/22585333
Gram:NERIST, Nirjuli

Email: @nerist.ernet.in
:@nerist.ac.in
Website: www.nerist.ac.in

DECLARATION

We, the students of Bachelor of Technology in the Department of Electrical Engineering, Session 2022–2026, North Eastern Regional Institute of Science and Technology (NERIST), hereby declare that the work presented in this project report entitled “*Modelling and Simulation of UPQC For Power Quality Improvement*” is the outcome of our own Bonafide effort and is correct to the best of our knowledge.

This work has been carried out with due regard to engineering ethics, and the results embodied in this project report have not been submitted to any other university or institution for the award of any degree or diploma.

Sanishisha Khyriemmuja
(D/23/EE/202)

Swarup Kar Chaudhuri
(D/22/EE/105)

Dapmain Papang
(D/22/EE/018)

Gracefully Syngkli
(D/22/EE/103)

ACKNOWLEDGEMENT

The satisfaction and sense of accomplishment that accompany the successful completion of any task would be incomplete without acknowledging those whose encouragement, guidance, and support made it possible. First and foremost, we would like to express our deep sense of gratitude and indebtedness to the College Management of the North Eastern Regional Institute of Science and Technology (NERIST) for their kind support and for granting us permission to utilize the facilities available in the institute.

We express our sincere gratitude to Prof. Narendranath S, Director, (NERIST)' for his invaluable encouragement, suggestions, and continuous support from the early stages of this project. His constant motivation and guidance have been a source of inspiration throughout the course of our work.

We are deeply grateful to Dr. Radak Blange, Head and Professor, Department of Electrical Engineering, for his valuable advice, supervision, and vital contributions whenever required during this research. We are truly privileged to have had the opportunity to work under such an experienced and respected academician. The time spent under his guidance will remain in our memories for years to come.

We would also like to extend our heartfelt thanks to our project supervisor Dr. M. Deben Singh, Associate Professor, Department of Electrical Engineering, for his constant guidance, encouragement, and support throughout this project. His insightful suggestions and valuable feedback have been instrumental in the successful completion of this work.

We are equally thankful to our Course Coordinator, Prof. T. Vimal Prakash Singh, Associate Professor, Department of Electrical Engineering, for his cooperation, advice, and valuable time. His involvement and constructive feedback have greatly enriched our learning experience.

Finally, we express our sincere gratitude to all the faculty members, staff, and friends of the Department of Electrical Engineering for their constant support, encouragement, and assistance during this project.

ABSTRACT

The rapid growth of power electronic devices and nonlinear loads in modern electrical distribution systems has significantly increased power quality disturbances, including voltage sags, swells, harmonics, flicker, and current imbalance. These issues lead to reduced efficiency, malfunction, and premature failure of sensitive electrical equipment, making power quality improvement a critical concern for utilities and consumers alike. In this context, the Unified Power Quality Conditioner (UPQC) has emerged as an effective custom power device capable of addressing both voltage- and current-related power quality problems simultaneously.

This project presents the modelling and simulation of a UPQC designed to enhance power quality in a distribution system. The UPQC consists of a series active power filter and a shunt active power filter connected through a common DC-link capacitor. The series converter is responsible for compensating voltage disturbances at the point of common coupling, while the shunt converter mitigates current harmonics, balances load currents, and maintains the DC-link voltage. A suitable control strategy is developed to coordinate the operation of both converters, ensuring accurate detection and compensation of power quality disturbances under dynamic operating conditions.

The complete UPQC model is implemented in the MATLAB/Simulink environment, where various power quality issues such as supply voltage distortion, voltage sag and swell conditions, and nonlinear load currents are introduced. The performance of the UPQC is analysed based on voltage regulation, harmonic reduction, and overall system stability. Simulation results demonstrate that the proposed UPQC model effectively restores the load voltage to its desired level and significantly reduces source current harmonics, resulting in improved power quality and enhanced system reliability. The study confirms that UPQC is a practical and efficient solution for meeting the growing power quality demands of modern distribution networks.

Keywords: DVR, UPDC, PQ, D-STATCOM, Distribution Network, Harmonics

CONTENTS

SL. NO	TITLE	PAGE NO.
1	Certificate	ii
2	Declaration	iii
3	Acknowledgement	iv
4	Abstract	v
5	Table of Contents	vi
6	List of Tables	vii
7	List of Figures	vii
A	1. Introduction	
	1.1 Introduction to the Project	1
	1.2 Background and Motivation	2
	1.3 Objective of the thesis	3
B	2. Literature Review of Unified Power Quality Conditioner	
	2.1 Literature Survey	4-5
	2.2 Control techniques of UPQC topologies	6-7
	2.3 UPQC in Service and Future Trends	8
C	3. Custom Power Devices	
	3.1 Introduction	9
	3.2 Core Components and Functionality in simulation	10-11
	3.3 Modern Solutions for Harmonic Problems	11-14
	3.4 Superiority of UPQC over other devices	14-15
E	4. Power Quality	
	4.1 Introduction	16
	4.2 Definition of Power Quality	16
	4.3 Power Quality Problems	16-18
	4.4 Classification of Power Quality Problems	19-24
	4.5 Power Systems Distortion and Problems	24-25
	4.6 Sources of Poor Power Quality	25-26
	4.7 Need of Power Quality	26
	4.8 Solutions to Power Quality Problems	27
	5. Project Design	
	5.1 Overview	28
	5.2 Structure of UPQC	29-31
	5.3 UPQC Control Strategy	31-33
F	6. Project Testing	34-38
G	7. Conclusion And Future Scope	
	7.1 Conclusion	39
	7.2 Future Scope	39-40
H	8. References	41
I	9. Appendices	42-44

List of Tables

Sl/No	Name of the table	Page No
1	Table 1: Simulink parameters and their specifications	35

List of Figures

Sl/No	Name of the figure	Page No
1	Basic configurations of a Unified Power Quality Conditioner.	29
2	Equivalent single-phase representation of UPQC.	31
3	P-Q block and inverse P-Q block in shunt active power filter	32
4	Controller circuit of series active power filter	33
5	Power system without UPQC	34
6	Waveforms of the voltage and current without UPQC at the source and the load respectively	34
7	THD value without UPQC	35
8	MATLAB simulation of power system with UPQC device	36
9	Voltage waveforms in a power system with UPQC	36
10	THD value with UPQC device	36
11	Output waveform of voltage sag mitigation	37
12	Output waveform of voltage swell mitigation	37
13	Simulink output for power quality parameter measurement	37
14	Source current waveform after mitigation	38
15	THD value of source current after mitigation	38

CHAPTER – 1

INTRODUCTION

1.1 INTRODUCTION TO THE PROJECT

The modern power distribution system must provide reliable and at the same time high-quality electricity. Modern electrical distribution systems often include large amounts of loads. Because of this, the requirement for energy demands from the network increase. The presence of heavy loads and dynamic loads leads to the fact that there are frequent fluctuations of voltage and currents, thus causing disturbances in form of voltage sags, swells, spikes, harmonics, and waveform's distortions. Disturbances have a detrimental effect on the performance, reliability, and service life of the equipment used in the system.

In recent times, power quality became one of the most pressing issues in relation to the power generation company and consumers. This phenomenon is associated with the widespread use of nonlinear loads, the existence of which contributes to the growth of power quality disturbances in modern systems. Nonlinear loads, such as power electronics converters, adjustable speed drives, rectifiers, computers, and other electronic loads cause significant harmonic distortion, an increase in reactive power consumption, a phase voltage imbalance, and overheating of electrical equipment.

Passive compensation methods such as LC Filters and capacitor banks were used traditionally for harmonics reduction and power factor correction. These methods are simple and economical; however, they have various limitations such as static nature of compensation, resonance problem, and poor performance during different loads. Due to the changes in loads experienced by today's distribution systems, traditional methods alone become insufficient for maintaining proper power quality requirements. Therefore, active power compensation methods and custom power devices were introduced to overcome the above limitations.

Custom power devices are the devices that are specifically manufactured for improving power quality in distribution systems. DVR, D-STATCOM, and UPQC are some of the custom power devices that are frequently used. The DVR device is primarily designed for solving the problems related to voltage disturbances such as sags, swells, and interruptions whereas the D-STATCOM device deals with the problems like current harmonics, reactive power, and load unbalance problems. But it becomes difficult to eliminate voltage and current disturbances at the same time using these two devices separately.

The Unified Power Quality Conditioner (UPQC) eliminates such drawbacks by integrating the capabilities of both DVR and D-STATCOM into an integrated solution. In its turn, UPQC is a system which includes two voltage source inverters coupled with a common DC link capacitor that consists of a series inverter and shunt inverter. While the series inverter generates compensating voltage for maintaining stable load voltage during disturbance, the shunt inverter performs tasks associated with harmonic mitigation and reactive power compensation as well as generation of sinusoidal source current. Thus, by placing UPQC at the Point of Common Coupling (PCC) of the distribution network, it becomes possible to resolve several issues of power quality that arise because of nonlinearities and fluctuations of load.

Using UPQC allows for considerable reduction of Total Harmonic Distortion (THD), improvement of power factor, voltage stability enhancement, and increase in reliability and efficiency of the whole distribution system. Consequently, UPQC became one of the most efficient solutions to providing high power quality.

1.2 BACKGROUND AND MOTIVATION

Electric power distribution is very important when considering the transfer of electrical power to the consumer side. There have been many changes that occurred over the last few decades in terms of the amount of increase in demand for electricity. Also, the complexity of the loads connected has increased greatly as well. Therefore, nowadays, the distribution network needs to provide electricity to many sensitive electronic/electricity-powered loads where the provision of good-quality electricity becomes an important issue. On the other hand, practical power distribution networks experience many disturbances in the form of voltage sags, voltage swells, spikes, harmonic distortion, flickers, and interruptions.

The growing use of nonlinear devices including variable frequency drives, rectifiers, inverters, switch mode power supplies, industrial controls, and computerized devices has increasingly emerged as a cause for poor power quality. The current drawn by nonlinear loads results in the generation of harmonic distortion, waveform distortion, voltage imbalance, and increased reactive power. The mentioned problems impact the operations of utilities as well as consumers in terms of increased losses, reduced efficiency, overheating of motors and transformers, malfunctioning of sensitive devices, and shortened lifespan of electrical components. Poor power quality in the industrial sector leads to disruption in the manufacturing process, inefficiencies, loss of data, and financial implications.

Passive Filters, Capacitor Banks and traditional compensation methods have been used in the past to overcome harmonic distortions and enhance the power factor. Though cost effective and easy to implement, these methods suffer from numerous drawbacks such as fixed compensation properties, resonance problems, inability to react quickly, and inadequate flexibility when dealing with variable loads. Modern power systems need faster and more flexible compensation schemes that can respond quickly to dynamic disturbances and operate satisfactorily even at different load levels.

To overcome the limitations associated with traditional compensation schemes, the idea of using custom power devices came into practice as a means to ensure better power quality at distribution level. DVR and D-STATCOM are some of these custom devices that have been used extensively for voltage and current compensations, respectively. However, individually, neither of these systems could be used simultaneously to address disturbances caused by voltage and current changes. Consequently, the idea of Unified Power Quality Conditioner (UPQC) was formulated.

The UPQC utilizes both series and shunt compensation connected through a common DC-link capacitor, enabling simultaneous compensation of voltage sags, swells, harmonics, reactive power, load imbalance, and transient disturbances. The series inverter injects compensating voltage into the distribution line to maintain a stable and balanced load voltage, while the shunt inverter compensates harmonic currents and reactive power demand, ensuring sinusoidal source current and improved power factor. This integrated approach provides a more reliable, flexible, and comprehensive solution compared to standalone compensators.

In addition to that, the incorporation of alternative energy sources such as the solar photovoltaic panel and wind energy converter in the distribution network has made things more complex and non-linear. The inclusion of electric vehicle chargers, the application of smart grids, and the adoption of distributed generation have further complicated the problem of power quality. In such circumstances, the role of UPQC becomes critical in ensuring the stability, reliability, and efficiency of the power network.

Thus, the motivation for utilizing the UPQC stems from the requirement of a high-efficiency, high-speed, and multi-functional device that can be used to improve the power quality of the

distribution network. The capability of the UPQC to deal with several disturbances at the same time, decrease the Total Harmonic Distortion (THD), improve power factor, regulate voltage, and improve reliability is what makes it one of the most attractive technologies for electrical power networks today.

1.3 OBJECTIVES OF THE THESIS

As discussed previously, the optimization of the ratings of the UPQC system has seldom been studied so far. This thesis mainly works on optimal design and control implementation of the UPQC system and aims to study power quality problems in modern distribution systems and evaluate UPQC as a complete solution.

The first objective is to review the existing UPQC topologies and techniques. Specifically, the UPQC topologies and techniques should be classified to provide a clear understanding of the state-of-the-art techniques in the UPQCs, and to discover the research gaps in this area. The power quality issues in a microgrid are important since all the renewable sources come with power electronic interfaces which generate undesirable voltage and current harmonics.

The second objective of this thesis will be to identify the main types of problems associated with power quality that need to be tackled by the UPQC device. Also, it is required to define the criteria for power quality standards of the grids. Such standards are used for the evaluation of operation performance of the proposed UPQC devices. As it can be seen from the construction of the UPQC which includes a series converter and a shunt converter, their implementation will increase the production cost of the device compared to the conventional power quality compensators. Hence, the commercialization of the UPQC device has not yet been made. It is necessary to find the ways how to minimize the ratings of the device without reducing its efficiency to maximize its operational performance and minimize its production costs. In the proposed thesis, there is a need to develop a search technique that would allow identifying the optimal ratings of series and shunt converters included in the UPQC device. Also, an appropriate control scheme should be created.

The study of current available techniques such as capacitors, filters, DVR, SVC, D-STATCOM, and ESS, together with their shortcomings, further stimulates interest in implementing this project. Hence, the general objective of this research work is to demonstrate that UPQC is a comprehensive technique for enhancing power quality, minimizing losses, and increasing reliability.

CHAPTER – 2

LITERATURE REVIEW OF UNIFIED POWER QUALITY CONDITIONER

2.1 LITERATURE SURVEY

Power quality improvement has emerged as a key focus area in modern power systems, particularly in smart grids and renewable energy-integrated networks. A variety of research studies have explored advanced control strategies, optimization algorithms, and intelligent solutions to improve the performance of Unified Power Quality Conditioner (UPQC) systems. This section presents a comprehensive review of related literature in sequential order, highlighting the evolution of UPQC-based techniques and their integration with smart grid technologies.

Sahoo et al. investigated power quality analysis in microgrids using a UPQC optimized by Particle Swarm Optimization (PSO). Their study demonstrated that PSO could efficiently tune the controller parameters to minimize Total Harmonic Distortion (THD) and enhance voltage regulation. The results confirmed the capability of UPQC to mitigate voltage sags, swells, and harmonic distortions in renewable-based microgrids. The optimization approach enhanced both steady-state and dynamic performance, laying the groundwork for intelligent power quality control mechanisms. Kumar and Kaur extended this idea by developing intelligent solutions for power quality improvement in smart renewable distributed generation networks. They employed machine learning-based control strategies to optimize power flow, voltage stability, and harmonic suppression. Their research highlighted how integrating AI algorithms with UPQC could ensure adaptive learning and self-tuning capabilities, improving system resilience in fluctuating renewable environments.

In a related domain, Vikas et al. proposed a hybrid Deep Belief Network and Harris Hawks Optimization method for intrusion detection in Wireless Sensor Networks. Although the work primarily addresses security challenges, the adaptive hybrid model concept is highly relevant to power systems where smart monitoring and fault detection are essential. This study indirectly supports the idea of using hybrid intelligent models for UPQC control and grid protection mechanisms.

Piklom et al. focused on reference signal generation for UPQC systems applied in heavy rail electrification networks. Their work introduced an effective reference signal generation method that improved the system's response under dynamic loading conditions, particularly in urban rail systems. The proposed model demonstrated robustness in compensating voltage imbalances and harmonics, ensuring smooth operation in high-demand electrical environments. Sharma and Kumar explored the role of Artificial Intelligence (AI) in enhancing data security and privacy in smart cities, emphasizing that AI-driven analytics and predictive models can also benefit smart grid reliability. Their findings emphasized integrating AI for real-time anomaly detection and predictive maintenance—key aspects for maintaining power quality and ensuring system integrity in smart grid infrastructures.

Behera et al. presented a comparative analysis of UPQC optimization using soft computing techniques, validating the potential of AI-based algorithms like fuzzy logic and genetic algorithms for performance tuning. Their work demonstrated that soft computing could effectively minimize THD and improve the voltage stability margin compared to conventional control methods.

Singh et al. proposed a Battery-Assisted Unified Power Quality Conditioner for tidal-driven seaport microgrids. The hybrid system utilized energy storage to enhance dynamic voltage regulation and improve energy utilization efficiency. The study provided insights into integrating energy storage systems with UPQC for sustainable microgrid operation and power quality management. In the field of network security, the study “A Comprehensive Analysis of Security Mechanisms and Threat Characterization in Mobile Ad Hoc Networks” identified multiple layers of protection required for distributed systems. Similarly, “Optimization of Graph Neural Networks for Real-Time Intrusion Detection” emphasizes data-driven modelling for anomaly detection. Both studies provide valuable methodologies that can be applied to intelligent grid monitoring and the cybersecurity aspects of UPQC-enabled smart grids. Ahmad and Ullah focused on UPQC-based power quality improvement in microgrids, emphasizing the dual role of series and shunt compensators in mitigating voltage disturbances and harmonic distortions. Their results confirmed significant improvement in power factor and voltage regulation, reinforcing the reliability of UPQC in grid-connected microgrids.

Lukka et al. developed an ANFIS-based intelligent UPQC for renewable energy microgrids, which utilized Adaptive Neuro-Fuzzy Inference Systems to dynamically adapt to load changes. Their model improved system response time and minimized voltage deviations, showing superior results compared to conventional PI and fuzzy controllers.

Finally, Farook et al. proposed a Synchronized UPQC for hybrid energy storage power management systems, addressing synchronization challenges in distributed hybrid sources, maintaining voltage and current stability during load transitions.

Today, in this modern world, power quality has become a great issue. Like many industries and for domestic use, we need a voltage and current free from all types of harmonics and unbalances. Due to problems in power quality, there is a development of many methods to improve power quality by using active power filters. The concept of power was introduced by N.G. Hingrani . Power electronics devices consist of a diode, thyristors, IGBT, and diodes. The active power filters are used to remove harmonics from current of load side and make supply current completely sinusoidal, and it also mitigate the problems of supply voltage imbalance i.e. voltage rise/dip and make voltages at load side balanced of equal magnitude. The active power filters can be combined and made to remove both problems due to voltage and current harmonics. There are a wide range of controlling techniques for active power filters. The reactive power theory was used to do simulation of three phase three wire lines which is valid for both transient and steady state. The physical meaning of instantaneous reactive power theory was described in. The instantaneous reactive power theory with the non-linear loads is described in. The DVR model is discussed for removal of all kinds of voltage related problems. Here the operating systems consist of PLL and Park's transformation is used for simulation. In three phase simulation of series active power filter is done for removal of voltages unbalances in supply side and make load voltage balanced and regulated. In chapter, the operation of DSTATCOM is explained. In the operation of three phase four wire shunt APF is explained which is used to suppress load current harmonics which is due to non-linear loads. As the power quality is the most important factor so to get improved power quality and removal of all type of harmonics from voltage and current we study UPQC which is a very versatile device and can be used for both mitigate the problems due to current harmonics and voltage disturbances. The voltage source inverter active filters are used for removal of power quality problems. The shunt active power filter is used to remove all the problems related to current harmonics and reactive power compensation so that the power quality will improve.

2.2 CONTROL TECHNIQUES OF UPQC TOPOLOGIES

The control unit is the most important part of the UPQC system. Rapid detection of disturbance signals with high accuracy, fast processing of the reference signal, and high dynamic response of the controller are the prime requirements for desired compensation. The series converter control ensures that the load voltage remains sinusoidal and undistorted by injecting the necessary compensating voltage. The supply voltage is measured and transformed into the d-q reference frame using Park's transformation, after which the reference voltage signals are generated. The deviation between the actual and reference voltage is processed using a Proportional-Integral (PI) controller, which regulates the Pulse Width Modulation (PWM) signals driving the inverter switches. The shunt converter, in turn, compensates for current-related distortions. It measures the load current, converts it into d-q coordinates, and extracts the harmonic and reactive components. The PI controller regulates the DC-link voltage and ensures that compensating currents are injected to maintain the source current sinusoidal and in phase with the voltage. This coordinated control between the series and shunt converters enables real-time compensation of both voltage and current disturbances, ensuring superior dynamic performance and improved power quality in the system. Control strategy plays the most significant role in any power electronics-based system. It is the control strategy which decides the behavior and desired operation of a particular system. The effectiveness of a UPQC system solely depends upon its control algorithm. The UPQC control strategy determines the reference signals (current and voltage) and, thus, decides the switching instants of inverter switches, such that the desired performance can be achieved. There are several control strategies/algorithms/techniques available in the existing literature that have successfully applied to UPQC systems.

Below are the different control strategies for a unified power quality conditioner to improve the power quality problems. Control strategy plays the most significant role in any power Electronics based system. It is the control strategy which decides the behavior and desired operation of a particular system. The effectiveness of a UPQC system solely depends upon its control algorithm. The UPQC control strategy determines the reference signals (current and voltage) and, thus, decides the switching instants of inverter switches, such that the desired performance can be achieved. There are several control strategies/techniques available in the existing paper that have successfully applied to UPQC systems.

2.2.1. Synchronous 'D-Q' Reference Base Theory

This algorithm relies on the Parks transformation where three-phase voltage and current signals are transformed into a synchronously rotating frame. The active and reactive components of the system are represented by the direct and quadrature component, respectively. In this approach, fundamental quantities become d-q quantities which can be separated easily through filtering. To implement the synchronous reference frame, some kind of synchronizing system PLL should be used. The system is very stable since the controller deals mainly with the d- q quantities. The computation is instantaneous but occurs time delays in filtering the d- q quantities. This method is applicable only for three-phase systems. The modified synchronous reference frame, named as the 'instantaneous id-iq method' is also proposed. This method is like synchronous reference frame method Except that the transformation angle is obtained from the voltage components. The speed referential is no longer constant, but it varies instantaneously depending on the waveform of the three-phase voltage system. In this method, no synchronizing circuit is needed.

2.2.2. Hysteresis Control Algorithm

In this control strategy, the unit vector templates are generated using PLL from three phase voltages measured at PCC. It generates the phase angle for the reference currents. The output of PLL together with a proper phase shift is used to generate three phase unity sinusoidal reference source currents. The magnitude of the reference source current is decided by the output of the PI controller. The PI controller compares the measured DC link voltage with the reference value, and its output is a measure of fundamental source current required to charge the capacitor. The measured source currents are compared with their reference values, and the switching pulses for the shunt APF are generated using a hysteresis controller.

2.2.3. 'P-Q-R' Instantaneous Power Theory

This method provides an analysis and control algorithm for a three phase four-wire Unified Power Quality Conditioner (UPQC) based on p-q-r instantaneous power theory. The p-q-r theory transforms a three phase four-wire voltage space vector into a single voltage and the corresponding currents into a based active power p-axis component and two imaginary power components, q-axis and r-axis. If there are harmonics and negative sequences in the voltage, the calculated reference current is not sinusoidal. In this method, an extra q-axis component is used to add to the original current compensation strategy based on p-q-r theory to maintain a sinusoidal current waveform under distorted voltage.

2.2.4. Switching Control Method

In this method, six single phase H-bridge inverters are used in the structure of UPQC connected to a common dc storage capacitor. Of these six inverters, three of them are used for series voltage insertion and the other three are used for shunt current injection. The UPQC current and voltage reference are generated based on fourier series extraction of fundamental sequence components using half cycle running(moving) averaging. They also propose a linear quadratic regulator-based switching controller scheme that tracks a reference using the proposed compensator. This method is suitable for both utilities and customers having sensitive loads. From the utility standpoint, it can make the current drawn balanced sinusoidal. To accomplish this, the voltage at the point of common coupling must be of similar nature and also must contain the same number of harmonics at the source. From the customer's point of view, the UPQC can provide balanced voltages to their equipment that is sensitive to voltage dips. At the same time, the UPQC also filters the current harmonics of the load. Therefore, the operation of UPQC is ideal from both viewpoints.

2.2.5. SPWM Based Control

The SPWM controller is used to operate the voltage source inverter in such a way that the difference between the inverter voltage and the line voltage is widely adjusted so that the shunt APF generates or absorbs reactive power. The measured three phase voltage is fed to PLL to detect the phase angle of voltages. The measured voltage is passed through a first order low pass filter to attenuate voltage transients. This signal is then compared with a reference voltage, and the voltage error is fed to a PI controller. The output of the PI controller is the angle representing the shift between the system voltage and the shunt inverter voltage required to adjust the voltage of the dc link capacitor. This angle combined with the signal from PLL becomes the voltage modulating signal. The phase angle from the PLL is multiplied by a carrier, whose frequency is 33 times the operating frequency to generate the triangular signal whose amplitude is fixed between the extremities of unity. The triangular carrier is compared with the voltage modulating signal to obtain the firing pulses for the shunt APF.

2.3 UPQC IN SERVICE AND FUTURE TRENDS

The Unified Power Quality Conditioner (UPQC) is a highly effective custom power device designed to address multiple power quality problems in electrical distribution systems simultaneously. In service, UPQC combines a series of active power filters and a shunt active power filter, interconnected through a common DC link capacitor, and operates in coordination to improve both supply-side and load-side power quality. The series converter injects a controlled compensating voltage in series with the supply to eliminate voltage-related disturbances such as sag, swell, voltage harmonics, unbalance, flicker, and short-duration transients. At the same time, the shunt converter injects compensating current into the system to cancel current harmonics caused by non-linear loads, supply reactive power demand, balance unbalanced load currents, and maintain a near-unity power factor at the source. This dual compensation capability ensures that sensitive loads receive a stable and distortion-free voltage while the utility experiences clean, sinusoidal current, leading to improved efficiency, reduced losses, and enhanced reliability of the power system.

Looking toward the future, UPQC technology is evolving to meet the demands of smart grids, renewable energy integration, and increasingly complex non-linear loads. UPQC technology is expected to play an even more critical role due to the rapid growth of renewable energy sources, electric vehicles, microgrids, and smart distribution networks. One major trend is the integration of energy storage systems, such as batteries and supercapacitors, into the UPQC DC link, which enables extended voltage support during deep sags, temporary outages, and fault conditions. Another significant development is the use of wide bandgap semiconductor devices like silicon carbide (SiC) and gallium nitride (GaN), which allow UPQC converters to operate at higher switching frequencies with lower losses, resulting in smaller passive components, higher power density, and improved overall efficiency.

Advanced and intelligent control strategies, including artificial intelligence-based controllers, adaptive control, and model predictive control, are also emerging to provide faster dynamic response, better disturbance prediction, and improved compensation accuracy under varying grid conditions. Furthermore, future UPQC systems are moving toward modular and scalable designs, such as modular multilevel converter (MMC)-based UPQC, which offer higher reliability, easier maintenance, and flexible capacity expansion. With the addition of IoT-based monitoring and communication, UPQC can support real-time supervision, remote diagnostics, and integration with smart grid management systems. These advancements position UPQC as a key enabling technology for ensuring high power quality, system stability, and efficient energy utilization in future smart grids and modern power distribution networks.

CHAPTER- 3

CUSTOM POWER DEVICES

3.1 INTRODUCTION

Custom power devices are power electronics-controlled devices meant for distribution systems, improving power quality by dealing with problems like sags, swells, harmonics, etc., for industrial customers. There are several types of Custom Power devices. Among them, some are Active Power Filters (APF), Surge Arresters (SA), Battery Energy Storage Systems (BESS), Super Conducting Magnetic Energy Systems (SMES), Static Electronic Tap Changers (SETC), Solid State Fault Current Limiter (SSFCL), Solid State Transfer Switches (SSTS), Static VAR Compensator (SVC), Distribution Series Capacitors (DSC), Dynamic Voltage Restorer (DVR), Distribution Static Synchronous Compensators (DSTATCOM), Uninterruptible Power Supplies (UPS), Unified Power Quality Conditioner (UPQC). A power conditioning device installed in the medium voltage distribution network of an electrical utility serves the objective of providing protection to either an entire plant, an entire feeder, or a block of customers or loads. Such devices will have voltage ratings of input and output of 1 KV to 38 KV, and the rating of loads will be more than 500 kVA. This technique of installing a power conditioning device at such a level is termed as custom power.

The classification of custom power devices can be done into two major categories:

1. Network configuring type,
2. Compensating type.

The **Network Configuring Type** of devices changes the configuration of the power system network for power quality enhancement. SSCL (Solid State Current Limiter), SSCB (Solid State Circuit Breaker), and SSTS (Solid State Transfer Switch) are the most representative in this category.

The **Compensating Type** devices are used for active filtering load balancing, power factor correction, and voltage regulation. The family of compensating devices includes DSTATCOM (Distribution Static compensator), DVR (Dynamic voltage restorer), and Unified power quality conditioner (UPQC). DSTATCOM is connected in shunt with the power system while DVR is a series connected device that injects a rapid series voltage to compensate for the supply voltage. UPQC is a combination of DSTATCOM and DVR. It injects a series voltage and shunt currents into the system.

Despite the many approaches used to reduce voltage sags and swells, employing a customized power apparatus is the most effective way to cater to different needs. Custom Power refers to the utilization of electronic power regulators in the distribution system to address various issues of power quality. It guarantees that consumers receive pre-specified levels of quality and reliability of power supply, which may incorporate either one or all of the following qualities: absence of interruptions, minimal phase imbalance, reduced harmonics in the load voltage, low flicker in the load voltage, tolerance to variations, magnitude and duration of over-voltage and under-voltages within certain limits, and poor power factor loads without any adverse effect on the terminal voltage.

3.2 CORE COMPONENTS AND FUNCTIONALITY IN SIMULATION

The simulation is structured to test how each filter handles specific power quality issues in a power system.

3.2.1 Active Filter (AF)

An Active Filter is a power electronic device used to improve power quality by removing harmonics, compensating reactive power, and balancing load currents or voltages. It uses switches like IGBTs/MOSFETs and a control system to inject compensating signals into the power system. Types of Active Filters:

- a) Shunt Active Filter
- b) Series Active Filter
- c) Hybrid Active Filter

Shunt Active Filter (SAF): The Shunt Active Filter (SAF), also known as the Parallel Active Filter (PAF), is the most used active filter in electrical power systems. It is connected in parallel with the nonlinear load at the Point of Common Coupling (PCC) and is mainly used for compensating current related power quality problems. In modern distribution systems, nonlinear loads such as rectifiers, variable frequency drives (VFDs), switched-mode power supplies, and inverters draw distorted and non-sinusoidal currents from the supply. These distorted currents generate harmonics, increase reactive power demand, reduce power factor, and degrade overall power quality.

The shunt active filter continuously monitors the load current and identifies harmonic and reactive current components present in the system. Using a Voltage Source Inverter (VSI) controlled by advanced digital control techniques, the filter generates compensating current equal in magnitude and opposite in phase to the harmonic current produced by the load. These compensating currents are injected into the distribution system, thereby cancelling the harmonic components and making the source current nearly sinusoidal and in phase with the supply voltage. In addition to harmonic mitigation, the shunt active filter also performs reactive power compensation and improves the power factor of the system. It reduces current distortion, minimizes electrical losses, prevents overheating of transformers and motors, and improves the efficiency and reliability of the distribution network. Due to its fast response and adaptive compensation capability, the shunt active filter is widely used in industrial plants, renewable energy systems, automation systems, and power-electronic-based applications.

Series Active Filter: The Series Active Filter is connected in series with the distribution line through an injection transformer and is mainly used for compensating voltage-related disturbances in the power system. The primary objective of the series active filter is to maintain a clean, balanced, and regulated voltage at the load terminals, even when disturbances are present in the supply voltage.

The series active filter continuously monitors the supply of voltage and detects disturbances such as voltage sags, voltage swells, harmonics, imbalance, and transients. When such disturbances occur, the filter injects an appropriate compensating voltage in series with the supply line to restore the load voltage to its desired sinusoidal waveform. This process ensures that sensitive equipment connected to the system receives stable and distortion-free voltage under all operating conditions.

Unlike the shunt active filter, which mainly compensates for current harmonics and reactive power, the series active filter focuses on improving voltage quality. It acts as a voltage regulator and harmonic isolator between the source and sensitive loads. The series active filters are particularly useful in systems supplying critical loads such as semiconductor manufacturing

units, hospitals, communication systems, and industrial automation equipment where voltage stability is essential.

Hybrid Active Filter: The Hybrid Active Filter combines the characteristics of both passive filters and active filters to achieve efficient and economical power quality improvement. In this configuration, passive filters are generally used to compensate for dominant low-order harmonics, while the active filter compensates higher-order harmonics and dynamic disturbances. This combined approach improves overall harmonic compensation performance while reducing the rating and cost of the active filter. The passive filter section handles fixed harmonic frequencies and provides reactive power support, whereas the active filter section offers dynamic and adaptive compensation under varying load conditions. As a result, the hybrid active filter overcomes many of the limitations associated with passive filters, such as resonance problems and fixed compensation characteristics, while also reducing the power requirement of the active filter.

Hybrid active filters are particularly suitable for high-power industrial systems where using a fully rated active filter may become costly and impractical. They are widely used in steel industries, large manufacturing plants, renewable energy integration systems, and power electronic converter applications. By combining the advantages of passive and active compensation techniques, hybrid active filters provide improved system efficiency, reduced harmonic distortion, and enhanced power quality at a comparatively lower cost.

Overall, active filters play an important role in maintaining power quality in modern electrical distribution systems. The shunt active filter mainly compensates current-related disturbances, the series active filter mitigates voltage-related disturbances, and the hybrid active filter provides a combined and economical solution for harmonic mitigation and reactive power compensation. These filters are extensively used in advanced custom power devices such as the Unified Power Quality Conditioner (UPQC) for comprehensive power quality improvement in modern distribution networks.

3.2.2 Lightning and Surge Arresters

Arresters are designed for lightning protection of transformers but are not sufficient for limiting voltage fluctuations to protect sensitive electronic control circuits. As custom power protection devices, surge arresters are engineered for specific applications ranging from high-voltage transmission lines (up to 1,100 kV) to low-voltage PCBs and residential service panels, with designs optimized for energy discharge capacity, leakage current minimization, and environmental durability. Unlike standard fuses or circuit breakers that open the circuit during faults, surge arresters clamp transient voltage spikes caused by lightning or switching operations without interrupting the normal power flow, offering a maintenance-free solution with a mean time between failure exceeding 1,000 years for high-voltage industrial units.

3.2.3. Battery Energy Storage Systems (BESS)

Battery Energy Storage Systems (BESS) are increasingly deployed as custom power devices to provide tailored, reliable energy solutions for diverse industrial, commercial, and specialized applications. Manufacturers like National Battery Supply and Custom Power design these systems to meet specific voltage, capacity, and form-factor requirements, often utilizing LiFePO₄ or high-density lithium-ion chemistries.

3.2.4. Uninterruptible power supply (UPS)

Uninterruptible power supply (UPS) is the conventional response to circumvent production interruption and outage costs. In UPS, the load has received the power from the source via two stage operations conversion (ac/dc) and inversion (dc/ac). During voltage dip or an interruption, the load voltage is made constant by energy, generated through battery. The performance of ups depends on energy storage capacity of battery. According To P. bowler UPS are needed where the power losses can cause economical lose greater than ups cost. For high power load financially, it is not suitable because of two conversions the maintenance cost of battery has become too high.

3.2.5. Solid State Current Limiter

A solid-state current limiter (SSCL) integrated into a Unified Power Quality Conditioner (UPQC) enhances power quality by mitigating high fault currents, reducing voltage sag, and preventing outages. The system combines quick-acting power electronic switches (like IGBTs/IGCTs) with impedance networks, allowing the UPQC to manage both normal power quality issues and severe fault conditions within milliseconds.

SSCLs detect fault currents instantly and insert impedance into the line to limit current magnitude. When added to a UPQC—which typically consists of series and shunt inverters for harmonic, reactive power, and voltage regulation—the combined system (FCL-UPQC or FCL-DVR) provides comprehensive protection against both disturbances and high-level faults. The SSFCL allows current to flow with minimal resistance. A fast-detecting control algorithm triggers solid-state switches (e.g., thyristors) to commutate current into a fault-limiting path (such as a reactor or resistive circuit).

3.2.6. Compensating power devices

The compensating custom power devices are used for active filtering, load balancing, and power factor improvement voltage regulating (sag/swell). These devices are mainly three types – static shunt compensator, series, and hybrid compensator. These are also called DSTATCOM, DVR, and UPQC, respectively.

Distribution Static Compensator (DSTATCOM): DSTATCOM is a Voltage source inverter (VSI) based static compensator device (STATCOM, FACTS controller) applied to maintain bus voltage sags at the required level by supplying or receiving of reactive power in the distribution system. It is connected in shunt with distribution feeder with the help of coupling transformer. The DSTATCOM consists of a VSI, dc energy storage device, an ac filter and coupling transformer. In the power circuit, VSI converts DC voltage into controllable ac voltage, synchronized by ac filter, and connected to the AC distribution line through coupling transformers. The DSTATCOM can also rely on and absorb active power, by using energy storage in a sufficient amount. The operating principle of DSTATCOM that continuously monitors the load voltages and currents determines the amount of compensation required by the distribution system for a verity of disturbances. In this scheme, the active power flow is controlled by the angle between the ac system and VSI voltages. The reactive power flow is controlled by the difference between the magnitudes of these voltages. The DSTATCOM operates in both current and voltage control modes.

Static Series Compensator: Commercially, static series compensator is known as Dynamic Voltage Restorer (DVR). It is a high-speed switching power electronic control device. The world's first DVR has been installed in the Dukey power distribution system to protect a sensitive textile customer from voltage dips on August 26, 1996. DVR is a series connected

custom power device, designed to inject dynamically controlled voltage in magnitude and phase into distribution line via coupling transformer to correct load voltage. It consists of an energy storage device, a boost converter (dc to dc), a voltage source inverter, ac filter and coupling transformer, connected in series. Here, the dc capacitor bank is used as an energy storage device, which is interfaced with a boost converter. The boost converter regulates the voltage across the dc link capacitor that is used as a common voltage source for the inverters. The inverter generates a compensating voltage, which is inserted into the distribution system through a series matching transformer. In the case of voltage reregulation, the DVR controllers generate a reference voltage and compare it with source voltage and inject synchronized voltage to maintain the load voltage constant. The energy storage device provides the required power to synchronized injected voltage. The ac filter overcomes the effects on winding of coupling transformer and switching losses of control signal generating techniques for VSI.

Unified Power Quality Compensator (UPQC): It is the combination of back-to-back connected shunt and series compensators through a common dc bus voltage. In this dc link storage capacitor is connected between two voltage source inverters for operating as combination of shunt and series compensator. It is a most flexible device, can suppress current in shunt and voltage in series simultaneously. It can balance the terminal voltage and eliminate negative sequence current components at the same time. Two types of UPQC are suggested in literature surveys. One is called Left- Shunt UPQC, and another is known as Right-Shunt UPQC. The overall performance of right-shunt UPQC is better than left-shunt UPQC. When UPQC is connected between two feeders, then called Interline Unified Power Quality Compensator (IUPQC).

3.3 MODERN SOLUTIONS FOR HARMONIC PROBLEMS

Modern solutions for harmonic problems represent a shift from static filtering to dynamic, intelligent, and system-wide mitigation strategies. As power systems integrate more renewable energy and non-linear loads like EV chargers, these "active" technologies are becoming the standard for maintaining grid stability and meeting standards like IEEE 519-2022.

Active Harmonic Filters: AHFs are the "gold standard" for modern industrial and commercial sites. Unlike traditional passive filters, they monitor the network in real-time and inject counter-harmonic currents that are 180° out of phase with the distortion, effectively "cancelling" the harmonics. They adapt instantly to changing loads, handling harmonics up to the 50th or 51st order. Modern AHFs also provide reactive power compensation (power factor correction) and load balancing. Many units, such as those from Schneider Electric or Sin excel, can be connected in parallel to increase filtering capacity as the facility grows.

Hybrid Mitigation Systems: Hybrid filters combine the cost-effectiveness of Passive Harmonic Filters with the precision of Active Harmonic Filters. The passive part handles fixed, lower-order harmonics (like the 5th or 7th), while the active part suppresses higher-order or fluctuating harmonics. This setup reduces the required rating (and thus the cost) of the active electronics compared to a fully active system.

Integrated Power Quality Conditioners: Solutions like the Unified Power Quality Conditioner (UPQC) integrate both series and shunt active filters to fix voltage-related

(sags/swells) and current-related (harmonics) issues simultaneously. These are ideal for critical infrastructure like hospitals or data centers.

In-Line Reactors: In-line reactors or choke is a simple solution to control harmonic distortion generated by adjustable speed drives. The solution is to come up with inserting a relatively small reactor, or choke, at the input of the drive. The inductance prevents the capacitor from being charged in a short time and forces the drive to draw current over a longer time and reduces the magnitude of the current with much less harmonic content while still delivering the same energy.

Hybrid Filters: Hybrid filter is a filter topology which combines the advantages of the passive and active filters. For this reason, it is considered the best solution to eliminate the harmonic currents from the grid. The principal reason for the use of hybrid filters is the development of power semiconductors like MOSFETs and IGBTs. Moreover, from an economic point of view, hybrid power filters allow for reducing the cost of APF. Hybrid power filters can be classified according to the number of elements used in the topology, the treated system (single phase, three phase three legs or four legs) and the used inverter type (current source inverter or voltage source inverter).

3.4 SUPERIORITY OF UPQC OVER OTHER DEVICES

The superiority of the **Unified Power Quality Conditioner (UPQC)** lies in its "all-in-one" architecture. While other devices typically solve only one side of the power quality equation (either voltage or current), the UPQC manages both simultaneously.

Some of the properties that makes UPQC more dominant than the other custom power devices are:

- a) Unified Control (Voltage + Current)
 - D-STATCOM: Only compensates for current-related issues (harmonics, reactive power).
 - DVR (Dynamic Voltage Restorer): Only compensates for voltage-related issues (sags, swells).
 - UPQC: Combines both. By integrating a series converter (acting like a DVR) and a shunt converter (acting a D-STATCOM) through a common DC link, it provides a comprehensive solution that makes standalone units redundant.
- b) Common DC Link Advantage
 - The UPQC uses a shared DC capacitor between its two converters. This allows for active power exchange:
 - The shunt converter can draw power from the grid to keep the DC link charged.
 - That stored energy can then be used by the series converter to "inject" voltage during a sag.
 - Other systems would require separate, bulky energy storage or large capacitor banks for each individual function.

c) Simultaneous Multi-Problem Mitigation

The UPQC is superior because it can handle a cocktail of problems at the same time:

- Harmonic Isolation: It prevents harmonics from the load side from reaching the source AND prevents source-side voltage harmonics from reaching sensitive loads.
- Flicker & Sag/Swell: It provides near-instantaneous voltage regulation.
- Power Factor Correction: It ensures the source current remains in phase with the source voltage.

Advantage of UPQC

- a) Simultaneous Compensation of Voltage and Current: The most significant advantage is that the UPQC can compensate for both supply-side voltage distortions (sags, swells, unbalance, flicker) and load-side current distortions (harmonics, reactive power) at the same time, maintaining a perfectly sinusoidal and regulated supply.
- b) Back-to-Back Converter Configuration: The series and shunt inverters are connected back-to-back through a common DC link, allowing them to exchange power to manage voltage sags or swells without needing an external energy storage system.
- c) Improved Power Factor: The shunt inverter component acts as a DSTATCOM to correct the load power factor to unity.
- d) Isolation of Sensitive Loads: The series inverter part acts as a DVR to protect sensitive loads from source disturbances, ensuring they operate properly, even with weak or distorted utility grids.
- e) Versatility in Applications: It is highly effective in applications such as electric arc furnaces and wind farms.
- f) Improved Dynamic Response: Compared to passive filters, the active nature of UPQC offers a much quicker response to transients and non-linear loads.
- g) Flexibility with Renewable Energy: UPQC can be integrated with Distributed Generation (DG) sources, such as solar PV or wind, where the DG can be connected to the DC link to compensate for voltage interruptions, making it superior to other FACTS devices.

CHAPTER – 4

POWER QUALITY

4.1 INTRODUCTION

The project was implemented by designing a distribution system model with UPQC connected at the Point of Common Coupling (PCC). A nonlinear load (such as rectifier/VFD-type load) was used to create real power quality problems like harmonics and imbalance. A suitable control strategy was developed to:

- Detect voltage and current disturbances
- Generate switching signals for inverters
- Maintain proper compensation

The system was modelled and simulated using software tool i.e. MATLAB/Simulink or equivalent. Different disturbances such as voltage sag, swell, harmonics, and load imbalance were introduced in the system. The UPQC responded in real-time by injecting compensating voltage (series part) and injecting compensating current (shunt part). System performance was monitored using parameters like:

- Voltage profile
- Current waveform
- Total Harmonic Distortion (THD)
- Power factor

Results before and after applying UPQC were compared to evaluate performance in the power system. The implementation confirmed that UPQC improves voltage stability, power factor, harmonic reduction and the overall system reliability.

4.2 DEFINITION OF POWER QUALITY

The Institute of Electrical and Electronic Engineers (IEEE) Standard IEEE1100 defines power quality as “The concept of powering and grounding sensitive electronic equipment in a manner suitable for the equipment.” As appropriate as this description might seem, the limitation of power quality to “sensitive electronic equipment” current might be subject to disagreement.

Power quality is the set of parameters defining the properties of the power supply as delivered to the user in normal operating conditions in terms of the continuity of voltage and voltage characteristics. It refers to the condition of electrical power in a system, characterized by the stability and purity of voltage, current, and frequency, such that electrical equipment operates efficiently, safely, and without performance degradation.

4.3 POWER QUALITY PROBLEMS

Power quality problems are widely observed in modern power distribution systems due to the increasing complexity of electrical networks and the growing use of nonlinear and power-electronic-based loads. In distribution lines, electrical power is often subjected to disturbances such as voltage sags, voltage swells, transients or spikes, harmonic distortion, voltage flicker, interruptions, and load unbalance. These problems are mainly triggered by large load

variations, switching operations, system faults, and external events such as lightning. As distribution systems serve a mix of industrial, commercial, and residential consumers, the impact of these disturbances has become more severe and widespread.

Another critical power quality issue is the presence of transients or voltage spikes, which are sudden, high-magnitude, short-duration disturbances. These transients are commonly caused by switching operations of large inductive loads, capacitor bank switching, or lightning strikes. High-energy spikes can damage semiconductor devices, disrupt communication systems, and lead to data loss. In distribution networks supplying heavy or fluctuating loads, such transients are more frequent, making protection and compensation essential.

Harmonic Distortion is one of the most serious power quality problems in modern distribution systems. Harmonics are generated by nonlinear loads such as rectifiers, inverters, variable frequency drives, switched-mode power supplies, and other power electronic devices. These harmonics distort the sinusoidal waveform of voltage and current, leading to overheating of transformers, motors, and cables, increased power losses, malfunction of protective devices, and poor system efficiency. The severity of harmonic distortion is commonly measured using Total Harmonic Distortion (THD), which indicates the level of waveform distortion relative to the fundamental component.

In addition to these issues, voltage flicker caused by rapid and repetitive voltage variations can result in noticeable fluctuations in lighting intensity, leading to discomfort and reduced visual performance. Interruptions, whether momentary or prolonged, completely disrupt power supply and can cause production downtime and data loss. Load unbalance, especially in three-phase systems, leads to unequal phase voltages and currents, resulting in excessive heating and reduced efficiency of rotating machines. Furthermore, poor power factor, often associated with reactive power demand and harmonic currents, reduces system capacity and increases operational costs.

These power quality problems tend to intensify during peak load conditions and in networks with a high penetration of nonlinear and renewable energy sources. Traditional mitigation techniques such as passive filters and conventional compensating equipment are often insufficient to handle multiple disturbances simultaneously under dynamic conditions. This growing challenge has created a strong need for advanced solutions such as custom power devices, particularly the Unified Power Quality Conditioner (UPQC), which can effectively mitigate voltage and current-related disturbances in a coordinated manner.

Steady-state deviations refer to continuous or sustained variations in electrical parameters that move away from their rated or nominal values during normal operating conditions of a power distribution system. Steady-state deviations persist long enough to influence the long-term performance and reliability of electrical equipment. These deviations are commonly observed in voltage magnitude, frequency, and waveform shape, particularly in distribution networks supplying fluctuating and nonlinear loads. One of the most frequently encountered steady-state deviations is voltage magnitude variation. Voltage sags occur when the RMS value of voltage temporarily drops below 90% of its nominal level for durations ranging from half a cycle to one minute. Such conditions are typically caused by sudden load increases, motor starting, or system faults. On the other hand, voltage swells are characterized by an increase in RMS voltage above 110% of the nominal value for a similar duration, often resulting from sudden load removal or improper voltage regulation. When these voltage variations persist for more than one minute, they are classified as undervoltage or overvoltage, respectively. Prolonged exposure to under- or overvoltage conditions can lead to overheating, insulation stress, and premature failure of electrical equipment.

Another important steady-state deviation is frequency variation, which occurs when the system frequency deviates from its nominal value of 50 Hz or 60 Hz. Frequency stability is directly related to the balance between power generation and load demand. Any mismatch between generated power and consumed power causes the frequency to shift, affecting the speed of motors, timing of electronic systems, and overall system stability. Although frequency variations are generally small in well-regulated grids, they can become significant during large disturbances or high penetration of renewable energy sources.

Another type of disturbance is the transient and dynamic issues are short-duration power quality disturbances that occur suddenly in power distribution systems and can have severe impacts on sensitive electrical and electronic equipment. These disturbances are typically caused by external events such as lightning strikes, switching operations, fault clearing, and sudden changes in load conditions. Although they last for a very brief period, their high magnitude and rapid nature make them particularly damaging to distribution networks.

Transients are classified into two main categories: impulsive transients and oscillatory transients. Impulsive transients are characterized by very fast rise times and extremely high voltage or current magnitudes. They are most caused by lightning strikes or electrostatic discharges and appear as sharp spikes in the voltage waveform. These high-energy spikes can puncture insulation, damage semiconductor devices, and lead to immediate failure of sensitive equipment. Oscillatory transients, on the other hand, occur due to switching actions such as capacitor bank switching, transformer energization, or switching of large inductive loads. These transients manifest as decaying oscillations superimposed on the voltage waveform and can cause repeated stress on system components. Interruptions represent another critical dynamic issue, where the supply voltage drops to zero for a short or extended duration. Interruptions may be momentary, temporary, or sustained, depending on their duration, and are usually caused by severe faults, equipment failure, or protective device operation. Even short interruptions can lead to production downtime, data loss, and malfunction of automated systems, particularly in industrial and commercial facilities. Voltage fluctuations, commonly known as *flicker*, are repetitive and rapid changes in voltage magnitude. These fluctuations are often caused by cyclic loads such as arc furnaces, welding machines, or large motors. Flicker is most noticeable in lighting systems, where it causes visible variations in brightness, leading to discomfort for occupants and reduced productivity. Persistent voltage fluctuations also indicate instability in the distribution network and poor load regulation. Voltage and current imbalance are another dynamic issue frequently observed in three-phase distribution systems. It occurs when loads are unevenly distributed across the phases, resulting in unequal phase voltages and currents. Imbalance increases losses, causes excessive heating in motors and transformers, and reduces the efficiency and lifespan of electrical equipment.

These transient and dynamic power quality issues are closely linked to distribution line disturbances during heavy load conditions, where sudden load changes intensify spikes, fluctuations, and imbalances. Traditional mitigation methods are often inadequate to address multiple dynamic disturbances simultaneously. Advanced compensation devices such as the Unified Power Quality Conditioner (UPQC) provide an effective solution by simultaneously compensating voltage and current disturbances. By filtering transients, stabilizing voltage, and balancing currents, the UPQC enhances system reliability and protects sensitive loads in modern distribution networks.

4.4 CLASSIFICATION OF POWER QUALITY PROBLEMS

Power quality problems are deviations in voltage, current, or frequency from ideal conditions that affect the performance and lifespan of electrical equipment. Due to the increasing use of nonlinear loads, power electronics, and renewable energy sources, these disturbances have become more frequent.

According to IEEE Std. 1159 and IEC standards, power quality problems are classified based on duration, magnitude, and waveform characteristics.

4.4.1. Short Duration Voltage Variation

Short duration voltage variations are temporary changes in RMS voltage lasting from a few cycles up to one minute. They include voltage sags, swells, and interruptions. These variations are common in distribution systems due to faults, load changes, or switching operations. Even though they last briefly, they can cause malfunction or shutdown of sensitive equipment. Fast compensation devices like the UPQC (Unified Power Quality Conditioner) are used to maintain stable voltage.

- a) Voltage Sag: Voltage sag is a temporary drop in voltage, where the RMS value falls to about 10%–90% of its normal level for a short duration (from half a cycle up to one minute). It usually occurs due to short-circuit faults, starting of large motors or transformers, or sudden increases in load demand. These sags can cause equipment to shut down or reset and may lead to data loss in sensitive devices. To handle this, the UPQC injects compensating voltage through its series inverter, helping maintain a stable voltage at the load.
- b) Voltage Swell: Voltage swell is the opposite of a sag, where the voltage rises above its normal level (typically 1.1 to 1.8 per unit) for a short duration. It is mainly caused by sudden load removal, ground faults, or capacitor switching. Voltage swells can stress insulation, cause overheating, and damage equipment. The UPQC mitigates this by injecting a counter-voltage to bring the voltage back to safe levels.
- c) Interruption: An interruption is a condition where the voltage drops to nearly zero (less than 0.1 per unit), resulting in a temporary or complete loss of power. Interruptions can be momentary (less than 3 seconds), temporary (3 seconds to 1 minute), or sustained (more than 1 minute). They are usually caused by faults, protective device operation, or equipment failure. Interruptions can lead to complete shutdown, data loss, and production stoppage. The UPQC helps during short interruptions by supplying power using energy stored in its DC-link capacitor, maintaining continuity for a brief period

4.4.2. Long-Duration Voltage Variation

Long-duration voltage variations are sustained deviations in RMS voltage that last more than one minute. Unlike short-duration disturbances, these are steady-state conditions caused by prolonged system issues. They mainly include:

- Overvoltage
- Undervoltage
- Sustained interruptions

These variations occur due to improper voltage regulation, long-term load changes, or system configuration issues. If not controlled, they can reduce efficiency, cause overheating, and degrade equipment performance.

- a) Overvoltage: Overvoltage is a condition in an electrical power system in which the RMS value of the voltage rises above 110% of its nominal value and remains at that elevated level for more than one minute. It is categorized as a sustained voltage disturbance and is an important aspect of power quality because it directly affects the safe and efficient operation of electrical equipment. This phenomenon generally occurs due to imbalances between generation and load or due to improper system control. One of the primary causes is the excessive injection of reactive power into the system, particularly from capacitor banks, which increases the voltage magnitude. Faulty operation of voltage regulators or tap-changing transformers can also lead to improper voltage control, resulting in overvoltage conditions. Additionally, when large loads are suddenly disconnected, the reduction in current demand causes a rise in system voltage. Poor or broken neutral connections may further contribute to voltage imbalance and abnormal voltage rise. In long transmission lines that are lightly loaded, the Ferranti effect becomes significant, where the receiving-end voltage increases due to the line's capacitance.

The effects of overvoltage can be severe and detrimental to the power system. It imposes excessive stress on the insulation of electrical equipment, leading to gradual deterioration or even sudden failure. Devices such as motors, transformers, and sensitive electronic equipment may experience malfunction or permanent damage. Prolonged exposure to higher voltage levels reduces the overall lifespan of equipment and increases maintenance requirements. Moreover, overvoltage conditions pose safety risks, including the possibility of electrical fires and hazards to human life.

To mitigate overvoltage, appropriate control and protective measures must be implemented. Voltage regulating devices and surge protection equipment are commonly used to maintain voltage within safe limits. Proper grounding and continuous system monitoring are essential for stable operation. In modern power systems, advanced power quality devices such as the Unified Power Quality Conditioner (UPQC) are employed. The UPQC operates by injecting a compensating voltage through its series inverter, thereby maintaining the load-side voltage at a desired level even under disturbed conditions.

- b) Undervoltage: Undervoltage is a condition where the RMS voltage falls below 90% of nominal value for more than one minute. Undervoltage is a condition in a power system where the RMS value of voltage falls below the prescribed nominal level for a sustained period. It commonly occurs when the demand for electrical power exceeds the system's ability to supply it adequately, leading to a drop in voltage at the load end. This condition is primarily associated with heavy loading of the system. During peak load conditions, excessive demand causes a significant voltage to drop across transmission and distribution lines. Long distribution feeders with high impedance further aggravate this issue, as more voltage is lost along the line before reaching the consumer. A lack of sufficient reactive power support also contributes to undervoltage, since reactive power is essential for maintaining voltage levels in the system. Additionally, overloaded transformers or improper tap settings can result in inadequate voltage regulation, causing the output voltage to fall below acceptable limits.

The effects of undervoltage are particularly noticeable in electrical machines and consumer equipment. Motors tend to draw higher current to maintain their output,

which leads to overheating and potential damage to windings. The efficiency of machines decreases, and in severe cases, motors may stall. Lighting systems exhibit reduced brightness, leading to dimming, while electronic devices may malfunction or operate unreliably due to insufficient supply voltage. Over time, these effects can degrade system performance and shorten the lifespan of equipment. To mitigate undervoltage, it is necessary to implement proper voltage control and compensation techniques. Voltage regulation equipment such as tap-changing transformers and automatic voltage regulators helps maintain voltage within acceptable limits. Reactive power compensation, through devices like capacitor banks or FACTS devices, supports the system in maintaining voltage stability. Advanced solutions such as the Unified Power Quality Conditioner (UPQC) provide effective control by restoring the voltage through its series inverter, while simultaneously supplying or absorbing reactive power through its shunt inverter to stabilize the system. Sustained Interruption.

- c) Sustained interruptions: A sustained interruption is a complete loss of supply voltage (< 0.1 pu) lasting more than one minute. It is a long-duration power quality disturbance and usually requires manual intervention to restore power. In distribution systems, these interruptions occur when permanent faults or severe conditions force isolation of a network section. Unlike short interruptions, they cannot be cleared automatically.

The causes of these types of interruptions can be due to:

- a. Permanent faults in the system
- b. Severe weather (storms, lightning)
- c. Damage to lines, cables, or transformers
- d. Tree contact or insulation failure
- e. Failure of automatic reclosing after fault

This can affect the distribution system in various ways like complete blackout, shutdown of industrial processes, data loss and communication failure, and serious risks in hospitals and critical systems.

There are different methods or ways to mitigate these interruptions namely:

- a. Backup systems: UPS, batteries, generators
- b. Proper maintenance and protection schemes
- c. UPQC provides only short ride-through support (limited by DC-link energy)
- d. For long interruptions, UPQC must be combined with energy storage systems

4.4.3. Transients

Transients are short-duration, high-magnitude disturbances in voltage or current, lasting from microseconds to milliseconds. They appear as sudden spikes or oscillations caused by switching, faults, or lightning. Due to their fast rise time and high amplitude (often 10–100 times nominal voltage), transients are highly destructive to electrical and electronic equipment. There are mainly two types of Transients namely the Impulsive Transients Sudden, unidirectional spikes (e.g., lightning) and the Oscillatory Transients Damped oscillations superimposed on waveform.

- a) Impulsive Transient: An impulsive transient is a sudden and very high voltage spike that occurs in one direction for a very short time (microseconds). It is usually caused by lightning strikes, switching operations, or sudden interruption of current in inductive circuits. These spikes can damage insulation, harm transformers and electronic devices,

and even cause failure of sensitive equipment. To reduce this effect, the UPQC helps by clamping the voltage spikes through its series inverter, protecting the connected loads.

- b) Oscillatory Transient: An oscillatory transient is a disturbance where the voltage or current oscillates back and forth and gradually decreases over time. It can be classified into high, medium, and low frequency based on its range. These transients are mainly caused by capacitor switching, transformer energization, or switching of inductive loads. They can lead to insulation stress, resonance problems, malfunction of protection systems, and long-term equipment damage. The UPQC mitigates these issues by filtering and damping the oscillations, ensuring a stable and smooth voltage supply.

4.4.4. Voltage Fluctuations

Voltage fluctuations are continuous, repetitive variations in RMS voltage, typically occurring in the frequency range of 0.1 Hz to 30 Hz. They are most noticeable as light flicker in lighting systems. The causes of these fluctuations can be due to rapidly varying loads (arc furnaces, welders, compressors, large motors), frequent starting and stopping of machines, weak or overloaded distribution systems, or due to loose connections and unbalanced loads.

Characteristics of these fluctuations can be cyclic or random and are measured using flicker indices: (i) P_{st} (short-term) ≤ 1.0 (ii) P_{lt} (long-term) ≤ 0.65 . Voltage fluctuations can have effects on motor speed variations and torque pulsations, overheating and increased losses and reduced equipment lifespan. The effect can also be light flickering causing discomfort and reduced productivity.

The different ways to mitigate these fluctuations are:

- a. Control fluctuating loads and improve system strength
- b. UPQC reduces voltage fluctuations by injecting compensating current through its shunt inverter, stabilizing voltage and reducing flicker

4.4.5. Voltage Imbalance

Voltage imbalance occurs in three-phase systems when phase voltages are unequal in magnitude or not separated by 120° . It is usually limited to $\leq 2\%$ for safe operation. These are caused by uneven distribution of single-phase loads, faulty capacitor banks or blown fuses, loose connections or unequal line impedances, or transformer tap mismatch or poor grounding. Voltage imbalance results in excessive heating in motors due to negative-sequence currents or it can cause torque pulsations, vibration, and noise, neutral overloading in 4-wire systems which altogether reduced the efficiency and equipment life.

The ways to mitigate voltage imbalance can be through proper load balancing and system maintenance. UPQC corrects imbalance by implementing shunt inverter that balances currents combined with a series inverter that corrects voltage imbalance.

4.4.6. Waveform Distortion

When nonlinear loads the kind that draw current in irregular pulses rather than smoothly connect to the grid, they distort that ideal waveform into something jagged and irregular. This steady-state deviation is what we call waveform distortion, and it's one of the more persistent headaches in power quality management. The standard way to quantify this distortion is through Total Harmonic Distortion (THD). Most power quality standards cap voltage THD at the Point of Common Coupling (PCC) at 5% or below but with today's grids increasingly

loaded with power electronics, drives, and switching equipment, staying within that limit is easier said than done.

Waveform distortion isn't a single phenomenon it shows up in several distinct forms. Harmonics are the most common culprit: these are frequency components at integer multiples of the fundamental (3rd, 5th, 7th harmonics, and so on) that ride on top of the main waveform and corrupt its shape. Interharmonics are a trickier variant non-integer frequency component that don't fit neatly into the harmonic spectrum and are harder to filter out. Then there's DC offset, where a small direct current component sneaks into what should be a purely AC waveform problematic especially for transformers, which can saturate under such conditions. Notching appears as sharp dips in the waveform caused by the switching transitions in power converters, while noise refers to the broad spread of high-frequency disturbances that switching devices throw off continuously.

The primary drivers of waveform distortion are nonlinear loads devices that don't draw current proportionally to the applied voltage. Rectifiers, inverters, switched-mode power supplies (SMPS), and variable speed drives all fall into this category. Arc furnaces and other heavy power electronic equipment are also significant contributors. Beyond the loads themselves, switching operations within the network and inadequate grounding practices can further worsen distortion levels. The effects of waveform distortion ripple across the entire electrical system. Transformers, motors, and cables experience increased losses and overheating, shortening their operational life. Neutral conductors can become dangerously overloaded particularly in three-phase systems with high third-harmonic currents and capacitor banks, which are sensitive to harmonic voltages, are prone to premature failure. Motors suffer in a different way: harmonic torques cause pulsations and vibration, increasing mechanical wear over time. On the sensitive equipment side, distorted waveforms can cause malfunctions, data corruption, and erratic behaviour in everything from medical devices to industrial controllers. Taken together, these effects translate to measurably reduced efficiency and shorter equipment lifespans across the board.

Mitigation starts with good system design minimizing harmonic-generating loads where possible and applying passive or active filters to clean up the waveform at the source or point of impact. For more comprehensive compensation, the Unified Power Quality Conditioner (UPQC) stands out as a particularly effective solution. It operates on two fronts simultaneously and offers a robust, dual-action defence against waveform distortion.

4.4.7. Harmonics

Harmonics are unwanted frequency components that are integer multiples of the fundamental frequency (e.g., 50 Hz to 100 Hz, 150 Hz, etc.). They distort the normal sinusoidal waveform and reduce power quality. The level of distortion is measured using Total Harmonic Distortion (THD). In modern systems, harmonics are mainly caused by nonlinear loads like drives, rectifiers, and electronic devices. The types of Harmonics are given as under:

- a) Odd harmonics (3rd, 5th, 7th) – most common
- b) Triplen harmonics (3rd, 9th, 15th) – accumulate in neutral
- c) Even harmonics – indicate faults or asymmetry
- d) Characteristic harmonics – depend on converter type

Harmonics can have effects like overheating of transformers, motors, and cables, neutral overloading, capacitor failure (resonance), and equipment malfunction and reduced efficiency.

4.4.8. Electrical Noise

Electrical noise is high-frequency, irregular disturbance added to normal voltage or current. Unlike harmonics, it is non-periodic and spreads over a wide frequency range (kHz to MHz). In modern systems with power electronic devices, electrical noise is common and can disturb communication, control circuits, and sensitive equipment. The types of electrical noise can be thermal noise, shot noise, flicker noise ($1/f$), and EMI (Electromagnetic Interference).

Electrical noises cause switching of motors, relays, and converters, arcing contacts and fluorescent lamps, Electrostatic discharge (ESD), and High-frequency switching devices (IGBTs, SCRs) and can have effect on communication and data errors, malfunction of PLCs and control systems, relay misoperation, equipment aging and insulation damage.

4.5 POWER SYSTEMS DISTORTION AND PROBLEMS

Power system distortions refer to deviations in electrical quantities especially voltage and current from their ideal sinusoidal form. In modern distribution networks, these distortions have become increasingly common due to the widespread use of nonlinear and power-electronic-based loads. When large amounts of power are distributed under such conditions, the system often experiences unwanted spikes, dips, and waveform irregularities, which directly relate to the motivation for deploying advanced compensation devices such as the Unified Power Quality Conditioner (UPQC).

Unlike traditional power systems dominated by linear loads, today's networks supply a mixture of rectifiers, inverters, variable speed drives, UPS systems, electronic lighting, and digital equipment. These devices do not draw current proportionally to the applied voltage, leading to distorted currents that interact with system impedances and distort the supply voltage itself. As a result, power quality deteriorates not only locally but also across connected feeders. Power system distortions take several forms, often occurring simultaneously. These distortions are often amplified by resonance conditions, weak system impedance, and heavy power distribution during peak load operation.

Importantly, studies and industrial experience show that up to 80% of power quality disturbances originate within consumer facilities, rather than from the utility supply. Improper wiring, poor grounding, defective equipment, and dense concentrations of nonlinear loads significantly worsen distortion levels.

Several commonly used devices in industrial and commercial facilities are major contributors to power system distortion:

- a) Electric motors, especially during starting, cause voltage sags and demand reactive power, lowering power factor.
- b) Variable speed drives (VSDs) improve efficiency but generate significant harmonic currents if not properly filtered.
- c) Lighting systems, including fluorescent and LED lighting, contribute to harmonics and low power factor.
- d) Transformers may introduce harmonics due to core saturation and consume reactive power.
- e) Capacitor banks, while intended for power factor correction, can resonate with harmonics and amplify distortion.
- f) UPS systems and office equipment based on power electronics draw highly non-sinusoidal currents.

The combined operation of these devices leads to widespread distortion throughout the distribution system.

The consequences of power system distortions are serious and multifaceted. They can cause increased electrical losses and reduced system efficiency, overheating of transformers, motors, neutral conductors, and cables, often requiring derating, capacitor failures due to resonance and excessive harmonic currents, maloperation of protective relays and control systems and so on so forth. In industrial environments, these issues can lead to production downtime, process instability, and financial losses.

Traditional mitigation methods such as passive filters, oversized equipment, or standalone compensators are often insufficient for today's dynamic and complex loads. The UPQC provides a comprehensive and adaptive solution by addressing both current- and voltage-related distortions simultaneously. Through this coordinated operation, the UPQC significantly improves power quality, system stability, and reliability, making it highly suitable for modern distribution networks experiencing severe power system distortions.

4.6 SOURCES OF POOR POWER QUALITY

In a power distribution system, maintaining good power quality is essential for reliable and efficient operation of electrical equipment. However, in practical conditions, the voltage and current waveforms often deviate from their ideal sinusoidal form. These deviations are mainly caused by various disturbances originating from both the utility side and the consumer (load) side. Such disturbances lead to issues like harmonics, voltage fluctuations, imbalance, and transient conditions.

The major sources responsible for poor power quality are explained below:

- a) Nonlinear Loads: Nonlinear loads are one of the primary causes of power quality problems in modern systems. Devices such as rectifiers, variable frequency drives (VFDs), inverters, and switched-mode power supplies do not draw current in a smooth sinusoidal manner. Instead, they draw distorted currents, which generate harmonics in the system. These harmonics distort the waveform of voltage and current, reducing overall power quality.
- b) Large Load Variations: Sudden changes in load demand significantly affect system voltage. The starting or stopping of heavy electrical equipment like motors, compressors, and arc furnaces results in abrupt current changes. This leads to voltage sags, swells, and flicker, which can disturb the operation of sensitive devices.
- c) Switching Operations: Frequent switching operations in power systems also contribute to disturbances. The switching of capacitor banks, transformers, or large loads produces transient voltages and current surges. These transients are short-duration but high-magnitude disturbances that can damage insulation and electronic components.
- d) Faults in the Power System: Electrical faults such as short circuits, line-to-ground faults, or equipment failures create severe disturbances in the system. These faults result in voltage dips, interruptions, and sometimes complete loss of supply, thereby affecting system stability and continuity of service.

- e) Lightning and Environmental Effects: Natural phenomena like lightning strikes introduce very high voltage surges into the system. Additionally, environmental factors such as storms, rain, and temperature variations can impact transmission and distribution lines, leading to transient disturbances and insulation failures.
- f) Poor Wiring and Grounding: Improper wiring practices and inadequate grounding systems are common sources of power quality issues. Loose connections, unbalanced loads, and faulty earthing can introduce electrical noise, voltage imbalance, and unreliable system performance.
- g) Overloaded Distribution System: When the demand for electrical power exceeds the capacity of the distribution system, it leads to overloading. This condition causes undervoltage, excessive heating of equipment, and increased losses, ultimately reducing efficiency and system life.
- h) Renewable Energy Integration: The increasing integration of renewable energy sources such as solar and wind introduces new challenges to power quality. These systems rely heavily on power electronic converters, which can generate harmonics. Moreover, their intermittent nature causes fluctuations in voltage and power levels, affecting system stability.

4.7 NEED OF POWER QUALITY

Power quality means maintaining proper voltage, frequency, and waveform so that electrical equipment works safely and efficiently. In modern systems with electronic and nonlinear loads, good power quality is essential.

There is an increased concern of power quality due to the following reasons:

- a) New-generation loads that use microprocessor and microcontroller-based 25 controls and power electronic devices, are more sensitive to power quality variations than that equipment used in the past.
- b) The demand for increased overall power system efficiency resulted in continued growth of devices such as high-efficiency adjustable-speed motor drives and shunt capacitors for power factor correction to reduce losses. This is resulting in increasing harmonic level on power systems and has many people concerned about the future impact on system capabilities.
- c) End users have an increased awareness of power quality issues. Utility customers are becoming better informed about such issues as interruptions, sags, and switching transients and are challenging the utilities to improve the quality of power delivered.
- d) Most of the networks are interconnected these days. Integrated processes mean that the failure of any component has much more important consequences

With the rise of VFDs, automation, data centers, and renewable energy, issues like sags, swells, harmonics, and flicker have become common. Without proper control, these lead to equipment failure and inefficiency.

Therefore, good power quality is necessary because of multiple reasons be it economically or for safety as this helps improves the overall performance of the electrical power system.

4.7 SOLUTIONS TO POWER QUALITY PROBLEMS

The mitigation of power quality problems can be achieved in two ways. It can be done from either customer side or utility side. First approach used is load conditioning and the other is line conditioning. Load conditioning ensures that the equipment is less sensitive to power disturbances. They are based on PWM converters and connected in shunt or in series to low and medium voltage distribution system. Series active power filters operate in conjunction with shunt passive filters in order to compensate load current harmonics. Series active power filters operate as a controllable voltage source whereas shunt active power filters operate as a controllable current source. Both of these schemes are implemented preferable with voltage source PWM inverters, with a dc bus having a reactive element such as a capacitor. However, with the restructuring of power sector, the line conditioning systems or utility side solutions will play a vital role in improving the inherent supply quality. Some of the effective and economic measures can be identified as following:

- a) Lightning and Surge Arresters
- b) Thyristor Based Static Switches
- c) Energy Storage Systems
- d) Electronic Tap Changing Transformer
- e) Harmonic Filters

CHAPTER – 5

PROJECT DESIGN

5.1 OVERVIEW

The project aims at developing a system for improving power quality based on the use of Unified Power Quality Conditioner (UPQC). With the rapid advancement in technology and industrial activities, there have been increased cases of nonlinear loads, power electronic converters, and sensitive electronic devices that cause disturbances including voltage sag, swells, harmonics, reactive power requirement, unbalance, and distortion in electrical distribution systems. The disturbance of power quality negatively impacts the efficiency of operation of the power system. Hence, to deal with these challenges, UPQC has been proposed as a solution since it can compensate voltage as well as current disturbances. The simulation model of the UPQC will be implemented in the MATLAB/Simulink environment to carry out a performance analysis of the UPQC device in various conditions. In this case, the distribution network consists of an AC supply source, nonlinear load that causes disturbances, and UPQC device which is connected at the PCC.

The UPQC comprises mostly of two Voltage Source Inverters (VSIs), which include the series inverter and the shunt inverter. They are interconnected using a common DC link capacitor that connects the two inverter stages back-to-back. The series inverter is connected in series to the distribution line using a transformer and is used to correct voltage irregularities such as voltage sag, swell, imbalance, spikes, and distortion. It generates a compensating voltage in the distribution system to ensure a balanced and regulated supply voltage. The shunt inverter is connected in parallel with the distribution system and acts as an active power filter. It is mainly used to correct current harmonics caused by nonlinear loads, reactive power correction, power factor correction, and maintenance of sinusoidal source current. Additionally, the shunt inverter controls the DC link capacitor voltage to facilitate appropriate energy transfer from one converter to another.

The DC link capacitor is used as an interface between the two converters and allows for energy sharing between the UPQC. The result is a combination of actions of both the inverters, making the UPQC a series and shunt active power filter. The UPQC's control mechanism constantly measures the system parameters like voltage and current levels at various stages of the distribution system. On detection of any kind of disturbances in the system, suitable control pulses based on PWM technique are generated to control the inverters' switching action. This technique offers very high-speed responses and accuracy of compensation. The design is tested using both balanced and unbalanced load scenarios, by incorporating rectifier based non-linear loads and resistive loads in the project to prove the effectiveness of the proposed design. The simulation results show that the UPQC provides an effective solution in reducing Total Harmonic Distortion (THD), increasing power factor, regulating load voltage level and reactive power compensation.

From all of the above findings, it can be concluded that the UPQC is an excellent solution for resolving power quality issues in electrical distribution networks working with dynamic and nonlinear load scenarios.

5.2 STRUCTURE OF UPQC

UPQC is a series combination of series and shunt active power filters sharing a common DC link. The two active power filters have different functions. Series filters are operated as a controlled voltage source to suppress and isolate voltage harmonics; at the same time shunt filters act as a controlled current source to compensate for the current harmonics. This paper presents complete simulation of the UPQC system. The basic configuration of the UPQC is presented in Figure 1.

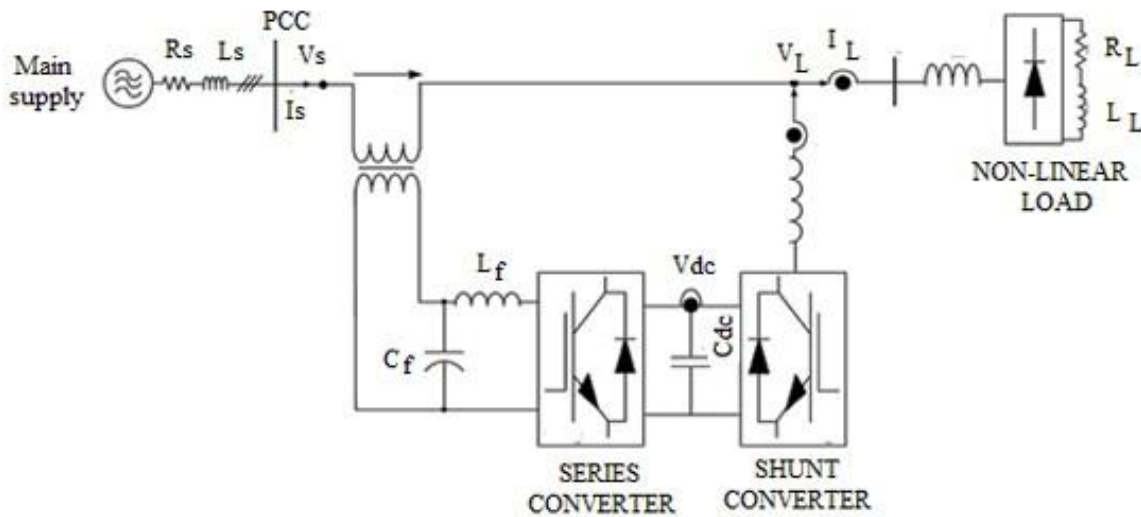


Figure 1: Basic configurations of a Unified Power Quality Conditioner.

Figure 1 shows a basic configuration of general UPQC consisting of two voltage source inverters, one act as a series APF and the other as shunt APF, which are connected back-to-back through the DC link capacitor. The series APF which is connected between the source and PCC using three single phase series transformers has the capability of compensating the voltage harmonics, voltage flicker, and improving voltage regulation. The shunt APF is connected through a small, rated capacity inductive filter to eliminate the high switching ripple content in the shunt APF injected current. Direct control strategy for UPQC in 3 phases, 4 wire system. The UPQC system is designed to address both voltage and current-related power quality issues simultaneously within a unified structure.

It consists of two Voltage Source Converters (VSCs): a Series Active Power Filter (SAPF) and a Shunt Active Power Filter (ShAPF), connected through a common DC-link capacitor. The series converter is connected in series with the grid through a coupling transformer and compensates for voltage disturbances such as sag, swell, and unbalance by injecting controlled voltages to maintain a constant load voltage. On the other hand, the shunt converter is connected in parallel with the load through an interfacing inductor and is responsible for current harmonics elimination, reactive power compensation, and maintaining unity power factor. The DC-link capacitor acts as an energy storage unit, balancing instantaneous active power between the converters. This dual compensating structure ensures that the supply voltage and load current remain sinusoidal and distortion free, significantly enhancing the overall power quality and system stability in smart grids.

5.2.1. Dc link capacitor

The two voltage source inverters are connected back-to-back through a DC Capacitor. DC Capacitor provides a DC voltage for the working of both the inverters. The DC capacitor also

provides a real power difference between source and load during the transient period and also acts as an energy storage element. During steady state, real power supplied by source should be equal to the sum of real power demand of load and a small amount of power which compensates for active filter. DC capacitor voltage should be equal to reference value but due to disturbance in real power balance between source and load due to change in load conditions, the DC capacitor value is changed from reference value.

5.2.2. Active Power filter

Traditionally passive filters were used for power quality improvement; the passive filters consist of a combination of capacitor, inductor, and resistor. Passive filters are used for harmonic filtering. Passive filters don't depend upon the external power source. It has many drawbacks such as being larger in size, resonance problems, the effect of source impedance on performance, and fixed compensation characteristics. So active power filters remove harmonics and do not have drawbacks such as passive filters.

Shunt Active Power Filter: Active power filters are devices which generate the same number of harmonics which are generated by load but at 180 phases shifted. Active power filters are devices such as amplifiers, etc. Shunt APF injects the compensating current in the line at the point of common coupling (PCC) so that the current at source sides become completely sinusoidal and free from distortions. Due to presence of non-linear load, there is harmonics & distortions in load current due to which source current also gets effected and source current becomes non- sinusoidal and distorted. So to remove this non-sinusoidal behavior of source current we use shunt APF which provides the compensating current which is same as harmonic generated by load but 180° phase shifted and this compensating current is given at PCC which helps in removing distortions from source current and makes source current completely sinusoidal . Shunt APF is also used for reactive power compensation, and it also removes all problems which arise due to current harmonics.

Series Active Power Filter: A series of active power filters are equipment which is used to mitigate the problems which are caused due to voltage distortions and voltage unbalance in source voltage. The voltage distortions and unbalance mean voltage dip, voltage rise, voltage fluctuations, voltage flicker; these are removed from the source voltage by means of series APF. A series of APF injects a voltage component in series with supply voltage and removes harmonic components. The series APF is used to remove all these voltage problems from supply voltage and make load voltage perfectly balanced and regulated. Series APF is connected in series with a transmission line with a series transformer. The turns ratio of series transformer should be proper so that the injected voltage should come properly.

5.2.3. Non-Linear Load

When the input current into the electrical equipment does not follow the impressed voltage across the equipment, then the equipment is said to have a nonlinear relationship between the input voltage and input current. All equipment's that employ some sort of rectification are examples of nonlinear loads. Nonlinear loads generate voltage and current harmonics that can have adverse effects on equipment designed for operation as linear loads. Transformers that bring power into an industrial environment are subject to higher heating losses due to harmonic generating sources (nonlinear loads) to which they are connected.

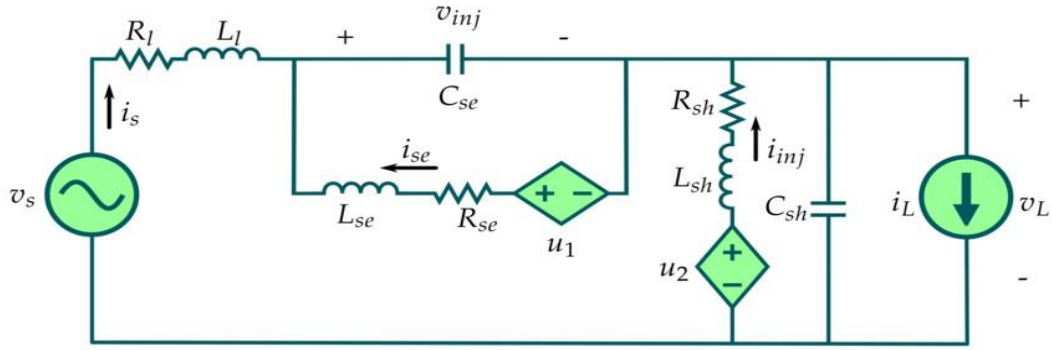


Figure 2: Equivalent single-phase representation of UPQC.

A Unified Power Quality Conditioner (UPQC) is a versatile custom power device that combines series and shunt active power filters (APFs) to mitigate both voltage and current-related power quality issues simultaneously. It is designed to provide clean, sinusoidal voltage to sensitive loads and ensure balanced, harmonic-free current from the source.

5.3 UPQC CONTROL STRATEGY

5.3.1. Shunt Control Strategy

Most APFs have been designed based on instantaneous active and reactive power theory (p-q), first proposed by Akagi et al in 1983. Initially, it was developed only for three-phase systems without neutral wire, being later worked by *Watanabe* and *Aredes* for three-phase four wires power systems. The method uses the transformation of distorted currents from three phase frame abc into bi-phase stationary frame $\alpha\beta$. The basic idea is that the harmonic currents caused by nonlinear loads in the power system can be compensated with other nonlinear controlled loads. The p-q theory is based on a set of 31 instantaneous powers defined in the time domain. The three-phase supply voltages (u_a, u_b, u_c) and currents (i_a, i_b, i_c) are transformed using the Clarke (or $\alpha\beta$) transformation into a different coordinate system yielding instantaneous active and reactive power components. This transformation may be viewed as a projection of the three-phase quantities onto a stationary two-axis reference frame. The Clarke transformation for the voltage variables is given by

$$\begin{bmatrix} u_\alpha \\ u_\beta \\ u_0 \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix} \quad \dots\dots\dots (i)$$

Similarly, this transform can be applied on the distorted load currents to give:

$$\begin{bmatrix} i_{l\alpha} \\ i_{l\beta} \\ i_{l0} \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} i_{la} \\ i_{lb} \\ i_{lc} \end{bmatrix} \quad \dots\dots\dots (ii)$$

The instantaneous active power $p(t)$ is defined by:

$$p(t) = u_a i_{la} + u_b i_{lb} + u_c i_{lc} \quad \dots\dots\dots(iii)$$

This expression can be given in the stationary frame by::

$$p(t) = u_{\alpha} i_{l\alpha} + u_{\beta} i_{l\beta}$$

$$p_o(t) = u_o i_{lo} \quad \dots\dots\dots (iv)$$

Where, p(t) is the instantaneous active power, p_o(t) is the instantaneous homo-polar sequence power. Similarly, the instantaneous reactive power can be given by:

$$q(t) = -1/\sqrt{3}[(u_a - u_b) i_{lc} + (u_b - u_c) i_{la} + (u_c - u_a) i_{lb}] = u_{\alpha} i_{l\beta} - u_{\beta} i_{l\alpha} \quad \dots\dots\dots (v)$$

It is important to notice that the instantaneous reactive power q(t) signify more than the simple reactive power. The instantaneous reactive power takes in consideration all the current and voltage harmonics, whereas the habitual reactive power considers just the fundamentals of current and voltage.

From equations (iv) and (v) the instantaneous active and reactive power can be given in matrix form by:

$$\begin{bmatrix} p \\ q \end{bmatrix} = \begin{bmatrix} u_{\alpha} & u_{\beta} \\ -u_{\beta} & u_{\alpha} \end{bmatrix} \begin{bmatrix} i_{l\alpha} \\ i_{l\beta} \end{bmatrix} \quad \dots\dots\dots (vi)$$

In general, each one of the active and reactive instantaneous power contains a direct component and an alternating component. The direct component of each present the power of the fundamentals of current and voltage. The alternating term is the power of the harmonics of currents and voltages. To separate the harmonics from the fundamentals of the load currents, it is enough to separate the direct term of the instantaneous power from the alternating one. A Low Pass Filter (LPF) with feed-forward effect can be used to accomplish this task.

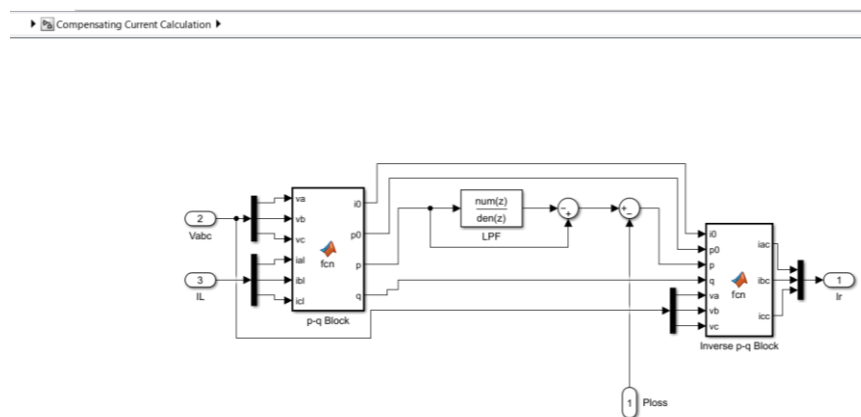


Figure 3: P-Q block and inverse P-Q block in shunt active power filter

Figure 3 shows the P-Q block used for generating the real and reactive power. It receives input Vabc and IL and based upon these two parameters the values i₀, p₀, p and q are calculated. The inverse p-q block is going to generate compensating current. The codes used in both function block is given in appendix C.

Therefore, by means of Clarke transformation and inverse Clarke transformation we generate the compensating current so it will act as a reference value for the next stage which is the hysteresis band controller.

5.3.2. Series Control Strategy

The series active power filter provides the voltage compensation. It generates the compensation voltage that synthesized by the PWM converter and inserted in series with the supply voltage, to force the voltage of PCC to become sinusoidal and balanced.

This power filter is connected via transformer in series with the distribution line. The load voltage is measured and is process by the controller. The controller has an inverter or simply an IGBT/diode which is used for generating the pulse for voltage source converter to mitigate the sag/swell and harmonic problems in the source voltage. In the controller, there consist of a block used for generating reference for the hysteresis controller. This reference is going to be compared with the load voltage after which it is process by hysteresis controller to generate the pulse for the voltage source inverter. Thus, the voltage source inverter will inject some voltage needed for the improvement of voltage in the power system at the load side. In the three phase inverter SCR conducts for 180° of a cycle. Thyristor pair in each arm are turned on with the time interval of 180° . The Simulink diagram of series controller is shown in the figure below (NOTE: V^* is connected to the 3-phase source inverter).

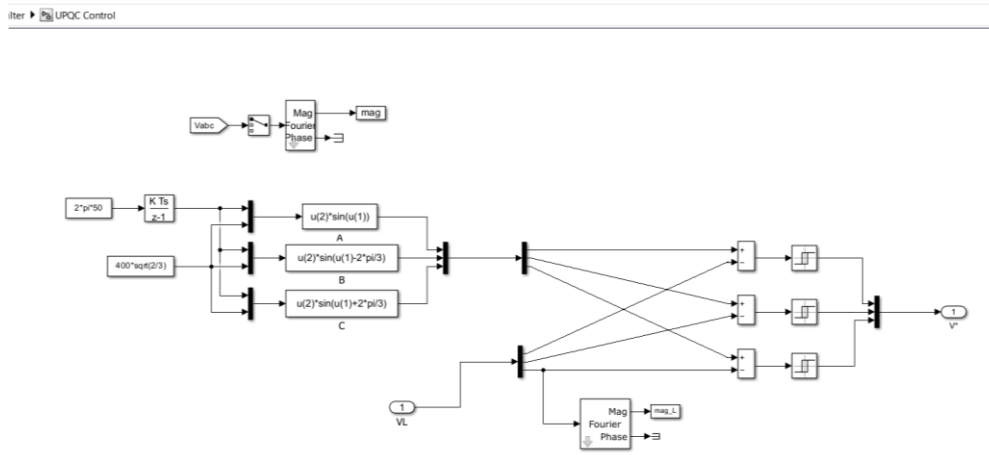


Figure 4: Controller circuit of series active power filter

CHAPTER – 6

PROJECT TESTING

Project testing is carried out to verify the performance of the Unified Power Quality Conditioner (UPQC) in improving power quality under different operating conditions. The main aim is to check how effectively the UPQC can reduce disturbances such as voltage sags, swells, harmonics, and load imbalance in a distribution system.

The testing of the project is performed using a simulated/experimental distribution system consisting of source (AC supply), nonlinear load (e.g., rectifier or variable load), UPQC (series and shunt converters with DC-link), measurement tools (voltage, current, THD analysis).

The Simulink model for power system UPQC has been simulated by using Matlab2023a. Table-1 gives the simulation parameters in detail. The effective comparative study on with and without using UPQC Device has been designed and presented as follows. In the circuit given in the figure below shows the system without using UPQC Device, the power quality problems occurred, further by using series active filter and shunt active filters it is used to maintain balanced and distortion free nominal voltage. The Simulink Diagram is shown in Figure 5.

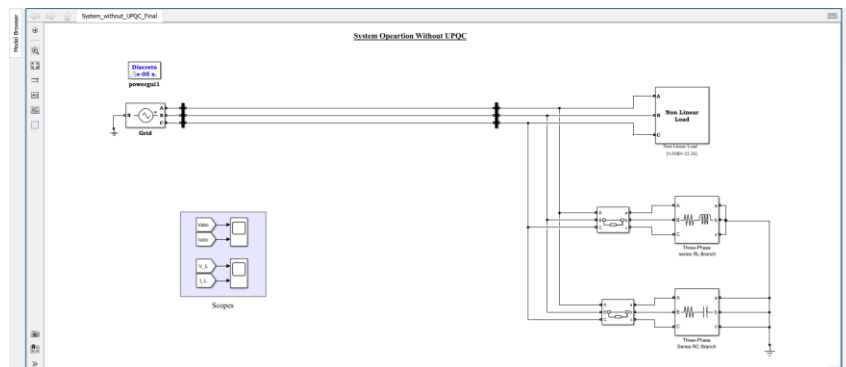


Figure 5: Power System without UPQC device.

The Power system without UPQC Device has been simulated and load voltage, load current obtained has been shown in Figure 6. We can see that the input and the output voltage contain some swell and sag due to the R-L and R-C load and the current is having some distortion.

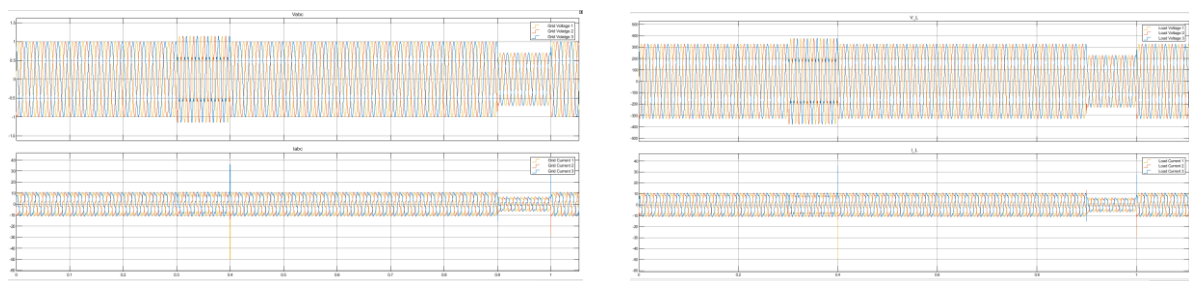


Figure 6: Waveforms of the voltage and current without UPQC at the source and the load side respectively.

Total Harmonic Distortion (THD) obtained the from power system without UPQC device is 22.22 % as shown in Figure 7.

This signifies that the system has significant harmonic distortions and is producing a highly distorted, non-sinusoidal waveform. As per the IEEE 519 standards, this value of THD is way above the recommended threshold which is 5 %.

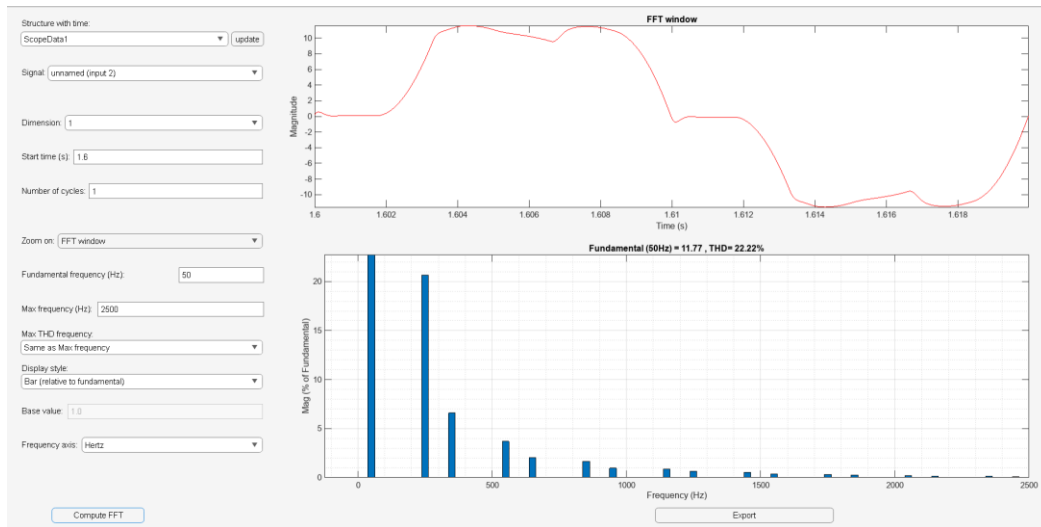


Figure 7: THD value without UPQC device

The table below gives the simulation parameters of the power system.

Table 1: Simulink parameters and their specifications

Parameters	Specification
Source voltages	Amplitude (V_{rms} ph-ph)= 400 Phase (deg)=0 Frequency (Hz)=50
Non-linear load	Three phase series RL branch: $R=0.4$ $L=15e-3$ Series RL load: $R=60$ $L=0.15e-3$
R-L load	Resistance(ohm)=20 Inductance(H)= $10e-3$
R-C load	Resistance(ohm)=3 Capacitance (F)= $15e-6$
Circuit Breakers	Switching time(s): [0.9 1] Breaker resistance(ohm) =0.01 Snubber resistance(ohm)= $1e6$ Snubber capacitance(F)=inf
DC link capacitor	$100e-6$ F
IGBT/Diode	Internal resistance(ohm)= $1e-3$ Snubber resistance(ohm)= $1e5$ Snubber capacitance(F)=inf
Transformer	Nominal power (VA)=4000 Frequency (Hz)=50

The Simulink model for the Power System using UPQC Device has been simulated by using MATLAB 2023a. Figure 8 shows Power Systems Using UPQC Device UPQC which consists of two inverters connected back-to-back and deals with both load current and supply voltage imperfections. UPQC can simultaneously act as shunt and series active power filters. It uses

balanced load with bridge rectifier and unbalanced load with RL load. The Power system using UPQC device has been simulated. It is used to maintain balanced and distortion free nominal voltage at the Load as shown in Figure 9.

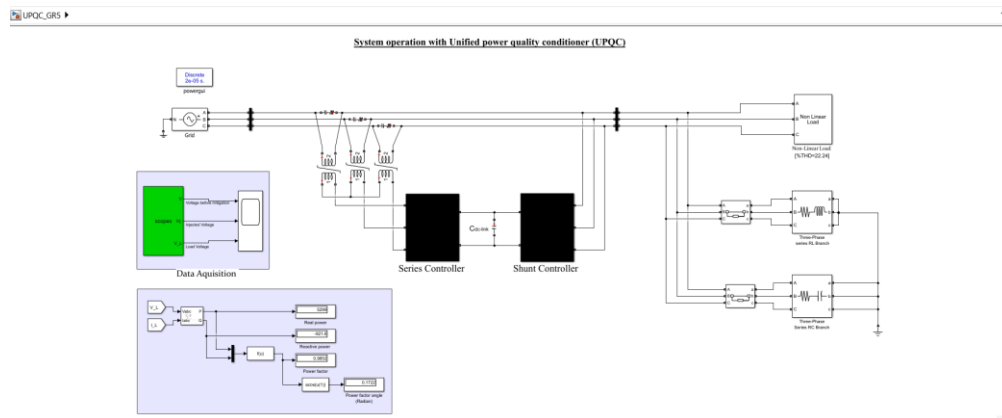


Figure 8: MATLAB simulation of Power system with UPQC device

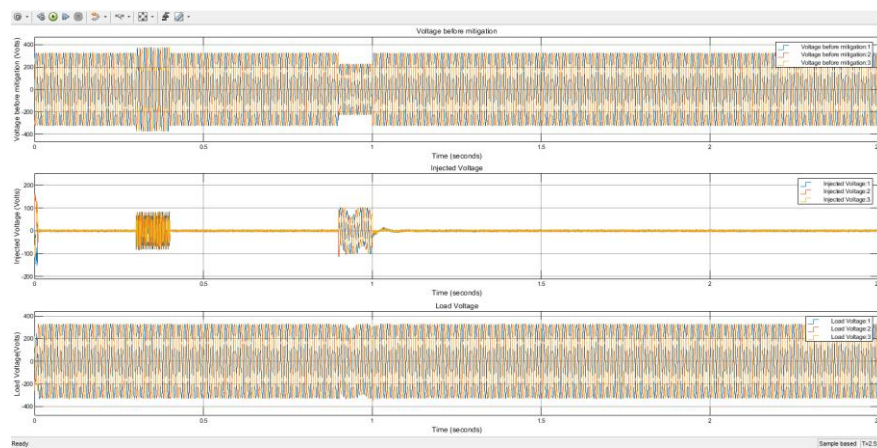


Figure 9: Voltage waveforms in a power system with UPQC

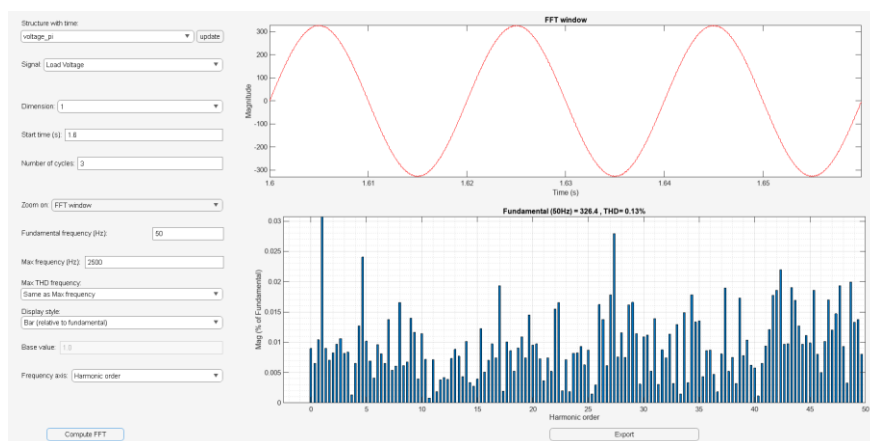


Figure 10: THD value with UPQC device

Different test conditions are applied to evaluate the performance of UPQC that is:

Voltage Sag Test: A sudden drop in supply voltage is introduced due to the R-L load. UPQC injects compensating voltage as shown in the figure 11. Upon injecting the voltage, the resulting load voltage remains stable.

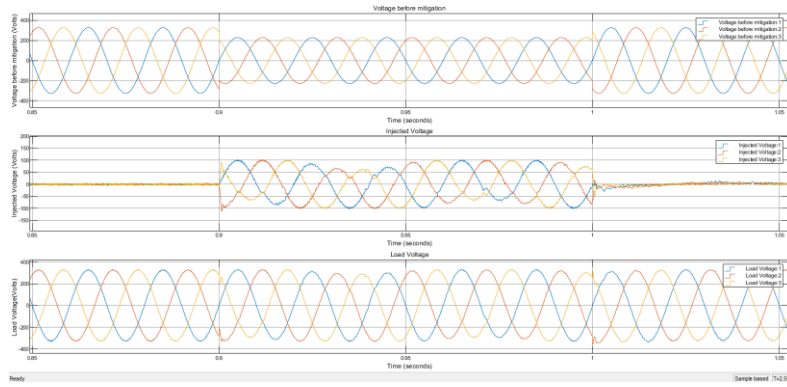


Figure 11: Output waveform of voltage sag mitigation

Voltage Swell Test: Supply voltage is increased above normal value due to the capacitive load at the rear end. UPQC reduces excess voltage through its series active power filter and hence the load voltage stays within limits.

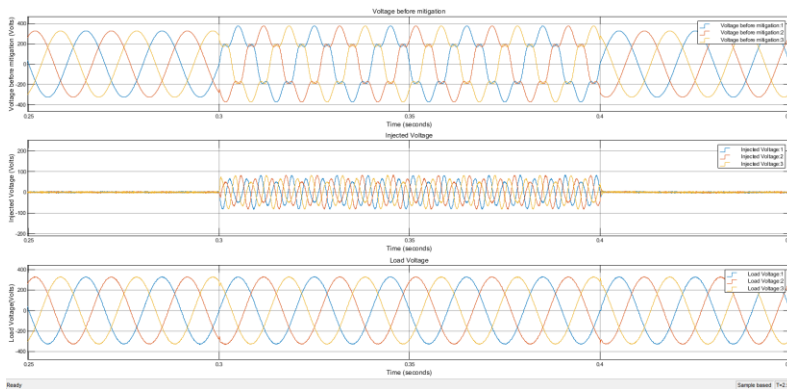


Figure 12: Output waveform of voltage swell mitigation

Harmonic Distortion Test: Nonlinear load is connected which results in harmonics generation in the power system. Using shunt active filter, it will act as a current source and injects an equal but opposite harmonic current into the line. Thereby the THD is reduced and waveform becomes sinusoidal.

Reactive Power Compensation: Load with low power factor is applied and UPQC supplies reactive power through the shunt controller. By taking over the reactive power supply, the UPQC ensures the source current stays in phase with the source voltage. As shown in figure 13 we see that the power factor obtained is 0.9852 indicating a highly effective compensation which improves the system operation and reducing energy losses.

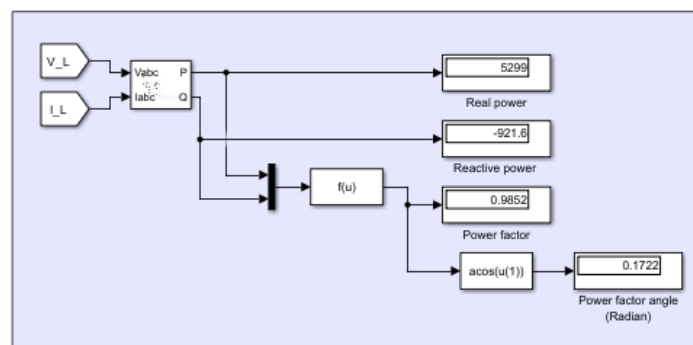


Figure 13: Simulink output for power quality parameter measurement

Output Current: The shunt active power filter improves the current at the source. The figure 14 shows that the source current follows sinusoidal shaped waveform and the THD value is reduced to 3.17% as shown in figure 15. Therefore, source current is improved as compared to the system operation without UPQC.

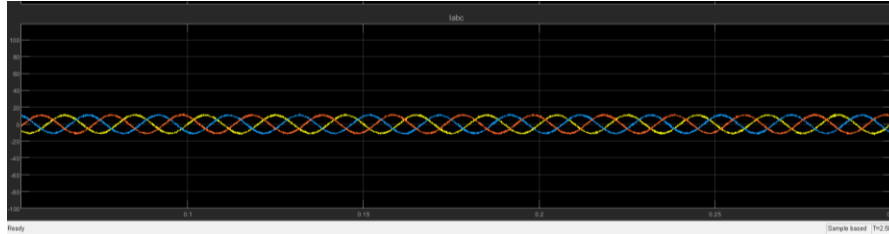


Figure 14: Source Current waveform after mitigation

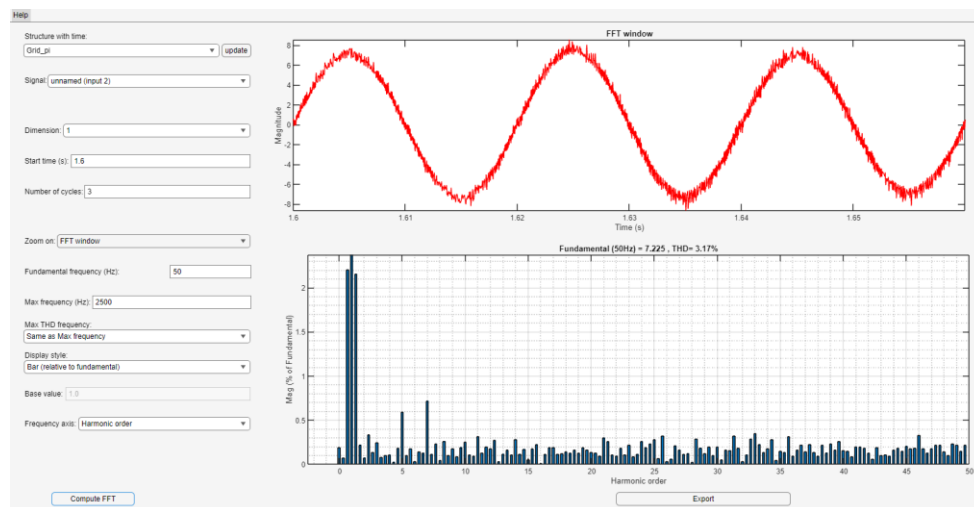


Figure 15: THD value of source current after mitigation

Overall, we can say that the performance parameters observed are the voltage profile (before and after compensation), current waveform quality, the Total Harmonic Distortion (THD), and power factor.

The testing results show that:

- UPQC effectively maintains constant load voltage
- Harmonics are significantly reduced
- Power factor is improved
- System becomes more stable and reliable

From the simulation results we mostly see that the UPQC device effectively mitigates power quality issues at the point of installation. However there exist some limitations and these are given as under:

1. The load voltage after compensation shows a minor oscillation or a “glitch” at time 0.9 to 1.0 seconds of the injection, which can be corrected by deploying modified PLLs.
2. Even if the source current gives a sinusoidal waveform, there is visible high frequency ripple that are likely due to switching frequency of power electronic circuits.
3. The simultaneous control of series and shunt controller requires sophisticated control techniques, which can be hard to implement.

CHAPTER – 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

This project focused on understanding power quality problems in modern distribution systems and evaluating the effectiveness of the Unified Power Quality Conditioner (UPQC) as a comprehensive solution. From the study, modern power systems face increasing issues such as voltage sags, swells, harmonics, transients, imbalance, and noise, mainly due to nonlinear loads, power electronic devices, and renewable energy integration. These disturbances negatively affect system performance, reduce efficiency, and shorten the lifespan of electrical equipment. Traditional solutions like capacitor banks, passive filters, DVR, and D-STATCOM can only address specific problems and are not sufficient for handling multiple disturbances simultaneously. In contrast, the UPQC provides a complete and integrated solution by combining series and shunt compensation.

The topology uses a capacitor in series with the interfacing inductor of the shunt active filter, and the system neutral is connected to the negative terminal of the dc-link capacitor to avoid the requirement of the fourth leg in the voltage source inverter (VSI) of the shunt active filter. The average switching frequency of the switches in the VSI also reduces. Consequently, the switching losses in the inverters reduce. The Main Merits of UPQC are harmonic elimination and simultaneous compensation of voltage and current, which improves the power quality offered for other harmonic sensitive loads. The performance of UPQC of different Controller at different operating conditions are analysed using MATLAB/Simulink Model. It had been analysed that the UPQC topology has less THD value in Load Voltage with reduced dc-link voltage as compared with the system without the UPQC device. The simulation can be extended to multi bus system.

Concluding, we can say that the UPQC proves to be an efficient, flexible, and reliable device for improving power quality in modern distribution systems, especially under dynamic and nonlinear load conditions.

7.2 FUTURE SCOPE

Although UPQC is highly effective, there are several areas where further improvements and research can be explored.

Integration with Energy Storage Systems (ESS): Enhancing UPQC with batteries or supercapacitors can provide support during long-duration interruptions and improve ride-through capability.

Advanced Control Techniques: Implementation of AI/ML-based controllers and intelligent algorithms can improve real-time performance and adaptability.

Renewable Energy Integration: UPQC can be further developed to handle high penetration of solar and wind energy, including bidirectional power flow in smart grids.

Cost Optimization: Research on reducing the cost of power electronic components and improving efficiency will make UPQC more accessible.

Hybrid Compensation Systems: Combining UPQC with other devices (like SVC or harmonic filters) can create more optimized and scalable solutions.

Application in Smart Grids and Microgrids: UPQC can play a major role in future smart distribution networks, ensuring stable and high-quality power supply.

The study confirms that UPQC is not just a solution for present power quality problems, but also a key technology for future power systems, capable of supporting modern electrical networks with higher efficiency, reliability, and flexibility. The proposed methodology not only enhances voltage profile stability but also achieves significant reductions in harmonic distortion, ensuring compliance with power quality standards under various operating conditions.

REFERENCES

1. Artificial Intelligence Based Three-Phase Unified Power Quality
<https://pdfs.semanticscholar.org/d5ab/4dd87a1023902123cfca40dbd6d74a88ba11.pdf>
2. Power Quality Improvement Using Unified Power Quality Conditioner (UPQC)
<https://ijeasm.com/PublishedPaper/3Vol/Issue4/2022IJEASM320221712-92269d27-70a7-4cb4-9028-b70477f510a222871.pdf>
3. Operation of Unified Power Quality Conditioner with Photovoltaic Arrays
file:///D:/Chrome/Reezm/Operation_of_Unified_Power_Quality_Conditioner_with_Photovoltaic_Arrays.pdf
4. Design of a Unified Power Quality Conditioner (UPQC)
<file:///D:/Chrome/Reezm/arastirmax-design-unified-power-quality-conditioner-upqc.pdf>
5. Solar PV Sustained Quasi Z-Source Network-Based Unified Power Quality Conditioner for Enhancement of Power Quality
<energies-13-02657-v2.pdf>
6. Power quality improvement using UPQC
<https://www.slideshare.net/slideshow/power-quality-improvement-using-upqc/143654575>
7. Simplified UPQC Control Method
<https://www.scribd.com/document/355096381/UPQC-1>
8. Multi-Objective Optimization Simulation of Unified Power Quality Conditioner (UPQC)
https://www.researchgate.net/figure/Equivalent-model-of-a-single-phase-grid-tied-UPQC_fig2_355472759
9. Modelling and Simulation of Unified Power Quality Conditioner (UPQC) for Power Quality Improvement in Smart Grids
<https://share.google/jCvsh3hmLlRQVE7GT>
10. UPQC Control for Power Quality Improvement through DQ0 and PQ Techniques
<https://www.youtube.com/watch?v=Zmp35AYKCjI>
11. Custom Power Devices for Power Quality Improvement
https://ijiset.com/vol7/v7s1/IJISSET_V7_I1_05.pdf
12. UPQC- File exchange-MATLAB central
<https://in.mathworks.com/matlabcentral/fileexchange/48497-upqc>

APPENDICES

The appendices section provides additional supporting information that complements the main content of this report. These details are included to improve clarity, provide reference material, and support the analysis and results presented in the study of power quality improvement using the Unified Power Quality Conditioner (UPQC).

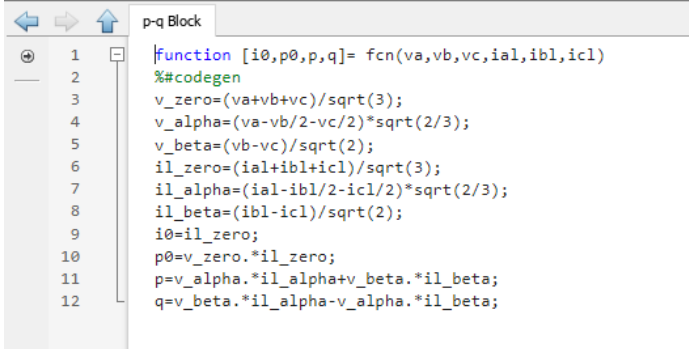
Appendix A: List of Abbreviations

- UPQC – Unified Power Quality Conditioner
- DVR – Dynamic Voltage Restorer
- D-STATCOM – Distribution Static Compensator
- SVC – Static VAR Compensator
- APF – Active Power Filter
- THD – Total Harmonic Distortion
- PCC – Point of Common Coupling
- VSI – Voltage Source Inverter
- IGBT – Insulated Gate Bipolar Transistor
- SCR – Silicon Controlled Rectifier
- ESS – Energy Storage System
- ETCT – Electronic Tap-Changing Transformer

Appendix B: Important Standards

- IEEE 519 – Recommended practices for harmonic control in power systems
- IEEE 1159 – Monitoring and classification of power quality disturbances
- IEC Standards – Guidelines for power quality limits and equipment performance

Appendix C:



```
p-q Block
function [i0,p0,p,q]= fcn(va,vb,vc,ial,ibl,icl)
%#codegen
v_zero=(va+vb+vc)/sqrt(3);
v_alpha=(va-vb/2-vc/2)*sqrt(2/3);
v_beta=(vb-vc)/sqrt(2);
il_zero=(ial+ibl+icl)/sqrt(3);
il_alpha=(ial-ibl/2-icl/2)*sqrt(2/3);
il_beta=(ibl-icl)/sqrt(2);
i0=il_zero;
p0=v_zero.*il_zero;
p=v_alpha.*il_alpha+v_beta.*il_beta;
q=v_beta.*il_alpha-v_alpha.*il_beta;
```

Function code for PQ block

```

function [iac,ibc,icc]= fcn(i0,p0,p,q,va,vb,vc)
%#codegen
v_zero=(va+vb+vc)/sqrt(3);
v_alpha=(va-vb/2-vc/2)*sqrt(2/3);
v_beta=(vb-vc)/sqrt(2);
i_zero=i0;
i_alpha=(v_alpha.*p+v_beta.*q)/(v_alpha.^2+v_beta.^2);
i_beta=(v_beta.*p-v_alpha.*q)/(v_alpha.^2+v_beta.^2);
iac=sqrt(2/3)*(i_zero/sqrt(2)+i_alpha);
ibc=sqrt(2/3)*(i_zero./sqrt(2)-i_alpha/2+sqrt(3)*i_beta/2);
icc=sqrt(2/3)*(i_zero./sqrt(2)-i_alpha/2-i_beta*sqrt(3)/2);

```

Function code for inverse PQ block

Appendix D: Key Equations

- Total Harmonic Distortion (THD):

$$THD = \frac{\sqrt{I_2^2 + I_3^2 + I_4^2 + \dots}}{I_1} \times 100\%$$

- Power Factor (PF):

$$PF = \cos \phi$$

- Reactive Power (Q):

$$Q = VI \sin \phi$$

Appendix E: Components of UPQC

- Series Inverter: Injects voltage to compensate sags, swells, and disturbances
- Shunt Inverter: Compensates current harmonics and reactive power
- DC-Link Capacitor: Connects both inverters and maintains energy balance
- Controller: Controls switching and compensation actions

Appendix F: Applications of UPQC

- Industrial plants and manufacturing units
- Data centers and IT infrastructure
- Hospitals and medical facilities
- Renewable energy integrated systems

Appendix G: Additional Notes

- UPQC provides simultaneous voltage and current compensation
- Improves power factor, reduces THD, and enhances system stability

- Suitable for modern smart grids and dynamic load conditions
- FFT Analyser: Utilizes the FFT function to compute the **Discrete Fourier Transform (DFT)** of a signal, converting time-domain data into the frequency domain to reveal its spectral components

Conclusion of Appendices

The appendices provide essential reference information that supports the understanding of power quality issues and highlights the effectiveness of UPQC as a comprehensive solution.